Lamina™



Flat Products

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Lamina™

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Lamina, Inc. began as a lamination die manufacturer in 1938. During the years from 1938 to the present Lamina has evolved into a manufacturer and distributor of die and mold components. Even though we no longer make dies, we do understand the die and mold maker's needs and objectives.

We have made it our business to supply quality products in a timely manner at a reasonable price. We know how to compete in the marketplace and supply you with the products you need when you need them.

With the introduction of this catalog featuring sliding guide components for molds, machinery and dies, we bring you a wider selection of wear and guiding products than in the past. It is our purpose to help you become more productive by helping to eliminate your downtime. With the products in stock, we will become your one-stop supplier and deliver the products you require each and every day.

Our catalog is presented in an easy-to-read format that will display the selection of products you need without a lot of extra data.

BRONZE-PLATED WEAR STRIPS & GIBS

Bronze-Plated Wear Strips & Gibs are made from low carbon extra-fine grain steel electroplated with a bronze alloy consisting of approximately 89% copper and 11% tin. The hardness is 180 BHn minimum. Lamina Bronze Plating has a natural lubricity and is ideal when used against hardened steel. We recommend any good grade of light machine oil suitable for machine way applications to extend the life of the wear surface.

The thickness of the bronze is .008 to .010". By using this thickness the bronze does not funnel out under heavy loads. The strength of steel backing keeps surface distortion to a minimum. Because most precision tool slide applications require either reworking or replacement if they develop so much as a .003 wear, Lamina Bronze thickness is more than adequate.

The superior wear properties of Lamina Bronze is not affected in working temperatures up to 400° F.

- Precision Machining on L-Gibs
- Precision ground sliding surface on wearstrips
- Close tolerances
- Consistent quality control
- Low Friction: Co-efficient of friction .2518
- Load factor 2700 PSI
- No galling or snow balling
- Magnetic - can be held on magnetic chuck
- Non-abrasive
- Non Seizing

BRONZE SELF-LUBRICATING WEAR STRIPS AND GIBS

Bronze Self-Lubricating wear strips and gibs are made from bronze with self-lubricating graphite plugs. These products can be used in applications up to 450° F (232°C).

Bronze is a strong material that resists wear, abrasion and deformation under high compressive loads.

Oil-impregnated graphite plugs cover 25-30% of the wear surface, providing a self-lubricating film of graphite on the wear surface.

Applying a film or mist of light machine oil on new self-lubricating parts before use, will enhance the self-lubricating properties. (Do not use grease on graphite-plugged surfaces.)

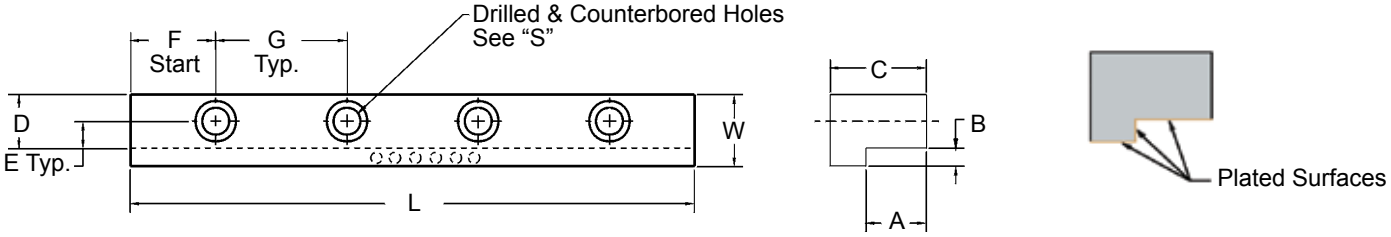
- Precision machining on L-Gibs
- Self Lubricating
- Precision ground wear surfaces on wearstrips
- Close tolerances
- Consistent quality control
- Resists abrasion
- Excellent wear characteristics

We would also like to remind you that these parts run best and last longest in a clean environment.

L-Gibs – Inch

Bronze-Plated Steel
With and without
graphite plugs



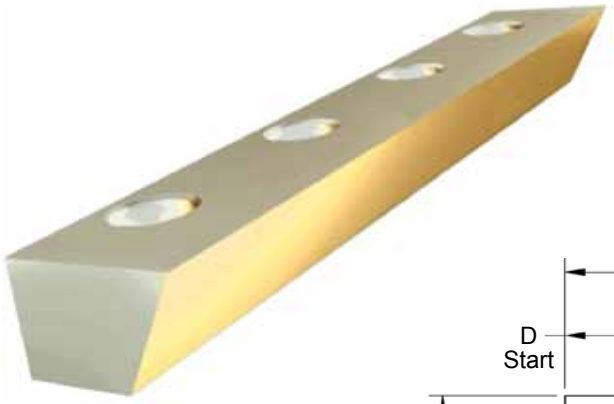


Standard L-Gibs with and without mounting holes
Self-Lubricating L-Gibs with and without mounting holes

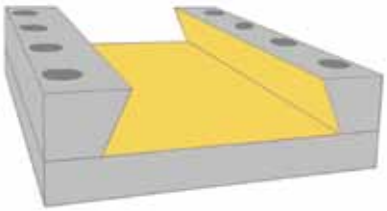
Hole Location ±.005
Unless noted, all dimensions ±.010

L-GIBS – INCH – BRONZE-PLATED – PLAIN & SELF-LUBRICATING												
PART NUMBER	PART NUMBER (Self-Lubricating)	L	W	A GH	B GW	C	D	E	F	G	S	
			+.000 –.005	+.001 –.000		+.000 –.005		±.005			Qty	Size
L-4-6-3	N/A	5.25	.75	.312	.188	.485	.562	.281	.875	1.750	3	1/4″
L-4-6-4	N/A	7.00									4	
L-4-6-5	SLL-4-6-5	8.75									5	
L-4-6-5-NH	SLL-4-6-NH	8.75						NO MOUNTING HOLES				
L-5-8-3	N/A	6.00	1.00	.375	.250	.610	.750	.375	1.000	2.000	3	5/16″
L-5-8-4	N/A	8.00									4	
L-5-8-5	SLL-5-8-5	10.00									5	
L-5-8-5-NH	SLL-5-8-NH	10.00						NO MOUNTING HOLES				
L-7-10-3	N/A	7.50	1.25	.500	.375	.860	.875	.437	1.250	2.500	3	3/8″
L-7-10-4	N/A	10.00									4	
L-7-10-5	SLL-7-10-5	12.50									5	
L-7-10-5-NH	SLL-7-10-NH	12.50						NO MOUNTING HOLES				
L-10-12-3	N/A	9.00	1.50	.750	.500	1.235	1.000	.500	1.500	3.000	3	3/8″
L-10-12-4	N/A	12.00									4	
L-10-12-5	SLL-10-12-5	15.00									5	
L-10-12-5-NH	SLL-10-12-NH	15.00						NO MOUNTING HOLES				
L-12-16-4	N/A	16.00	2.00	.875	.625	1.470	1.375	.687	2.000	4.000	4	1/2″
L-12-16-6	SLL-12-16-6	24.00									6	
L-12-16-6-NH	SLL-12-16-NH	24.00						NO MOUNTING HOLES				
L-16-20-6	N/A	24.00	2.50	1.250	.750	1.970	1.750	.875	2.000	4.000	6	5/8″
L-16-20-8	SLL-16-20-8	32.00									8	
L-16-20-8-NH	SLL-16-20-NH	32.00						NO MOUNTING HOLES				
L-20-24-5	N/A	30.00	3.00	1.500	1.000	2.470	2.000	1.000	3.000	6.000	5	5/8″
L-20-24-8	SLL-20-24-8	48.00									8	
L-20-24-8-NH	SLL-20-24-NH	48.00						NO MOUNTING HOLES				

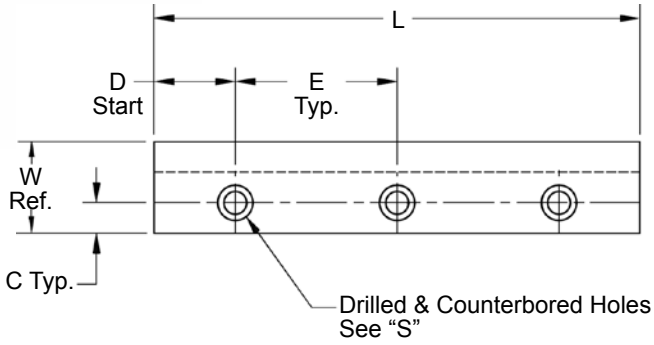
V-Gibs –Inch 60°

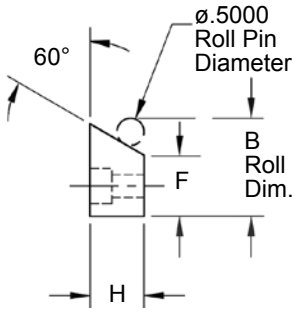


Bronze-Plated Steel



Use Lamina V-Gibs and Wear Strips right from stock to create assemblies.





Only Lamina stocks standard V-Gibs

Hole location ±.005
Unless noted, all dimensions ±.010

V-GIBS – BRONZE PLATED

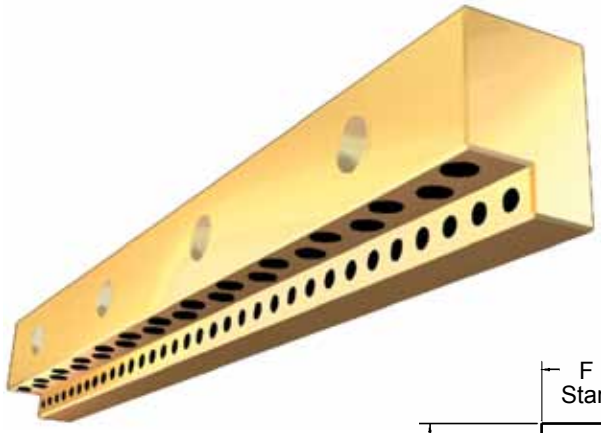
PART NUMBER	L	W Ref.	H	F Base Width	B Roll Dim.	C	D	E	S	
			+.000 –.005	+.000 –.001	+.000 –.001	+.000 –.005			No. Holes	Size
V-5-7-3	5.25	.92	.610	.562	1.2455	.281	.875	1.750	3	1/4″
V-5-7-4	7.00								4	
V-5-7-5	8.75								5	
V-6-10-3	6.00	1.18	.735	.750	1.4330	.375	1.000	2.000	3	5/16″
V-6-10-4	8.00								4	
V-6-10-5	10.00								5	
V-7-11-3	7.50	1.38	.860	.875	1.5580	.437	1.250	2.500	3	3/8″
V-7-11-4	10.00								4	
V-7-11-5	12.50								5	
V-8-12-3	9.00	1.70	.985	1.125	1.8080	.562	1.500	3.000	3	3/8″
V-8-12-4	12.00								4	
V-8-12-5	15.00								5	
V-10-16-4	16.00	1.97	1.235	1.250	1.9330	.625	2.000	4.000	4	1/2″
V-10-16-6	24.00								6	
V-12-20-6	24.00	2.49	1.470	1.625	2.3080	.812	2.000	4.000	6	5/8″
V-12-20-8	32.00								8	
V-14-24-5	30.00	3.01	1.720	2.000	2.6830	1.000	3.000	6.000	5	5/8″
V-14-24-8	48.00								8	
V-16-32-8	48.00	3.90	1.970	2.750	3.4330	1.375	3.000	6.000	8	3/4″

Lamina Bronze electroplated on steel has achieved universal acceptance for wear applications. The Lamina principle of using bronze for wear surfaces and steel for strength is seen on many products throughout the Lamina product line.

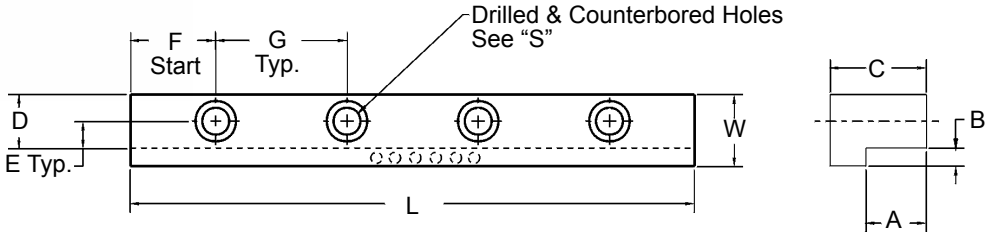
- **Plating thickness: .008 – .010**
- **One-piece construction with pre-drilled and counter-bored screw holes, machined complete.**

L-Gib – Inch

- Plastics Injection Molds
- Special Machines
- Press Gibbing
- Special Slide Applications



Bronze with
Self-Lubricating
Graphite Plugs



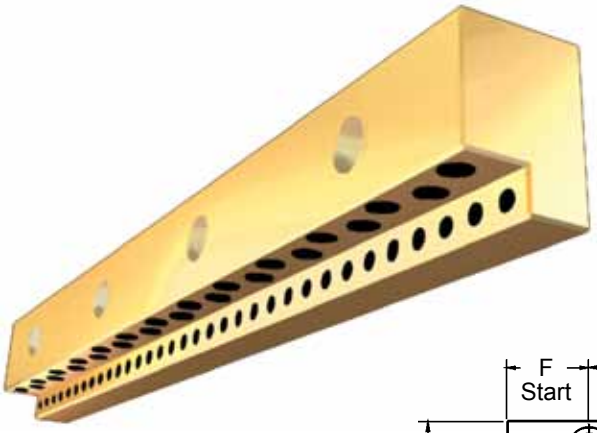
For L-Gibs without mounting holes use "N" at the end of the part number.

Unless noted, all dimensions ±.010

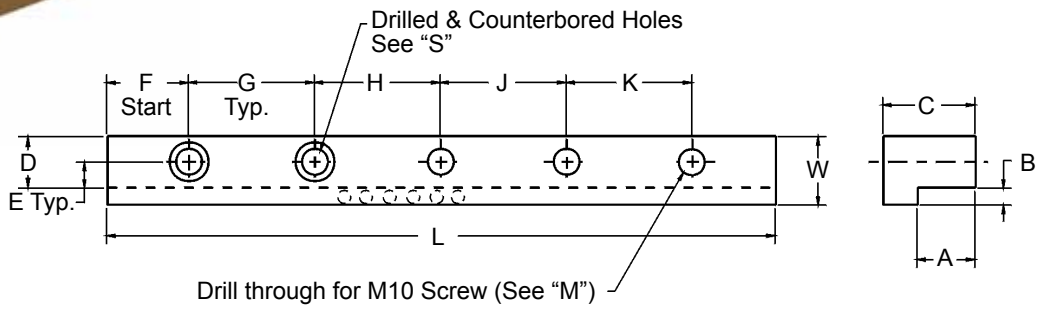
L-GIBS – INCH – SOLID BRONZE – SELF-LUBRICATING											
PART NUMBER	L	W	A GH	B GW	C	D	E	F	G	M	
		+.000 –.005	+.001 –.000		+.000 –.005		±.005			No. Holes	Size
L2LG42G	5.25	.75	.312	.188	.75	.56	.281	.875	1.750	3	1/4″
L2LG56G	7.00							.875		4	
L2LG70G	8.75							.875		5	
L3LG48G	6.00	1.00	.375	.250	.75	.75	.375	1.000	2.000	3	5/16″
L3LG64G	8.00							1.000		4	
L3LG80G	10.00							1.000		5	
L4LG48G	6.00	1.25	.500	.375	.88	.87	.437	1.000	2.000	3	3/8″
L4LG60G	7.50							1.250	2.500	3	
L4LG80G	10.00							1.250		4	
L4LG100G	12.50							1.250		5	
L6LG48G	6.00	1.50	.750	.500	1.25	1.00	.500	1.000	2.000	3	3/8″
L6LG72G	9.00							1.500	3.000	3	
L6LG96G	12.00							1.500		4	
L6LG120G	15.00							1.500		5	
L7LG64G	8.00	2.00	.875	.625	1.50	1.37	.687	1.000	2.000	4	1/2″
L7LG96G	12.00							1.500	3.000	4	
L7LG128G	16.00							2.000	4.000	4	
L10LG96G	12.00	2.50	1.250	.750	2.00	1.75	.875	1.500	3.000	4	5/8″
L10LG144G	18.00							2.250	4.500	4	
L10LG192G	24.00							3.000	6.000	4	
L12LG96G	12.00	3.00	1.500	1.000	2.50	2.00	1.000	1.500	3.000	4	5/8″
L12LG144G	18.00							2.250	4.500	4	
L12LG192G	24.00							3.000	6.000	4	

L-Gib – Metric

- Plastics Injection Molds
- Special Machines
- Press Gibbing
- Special Slide Applications



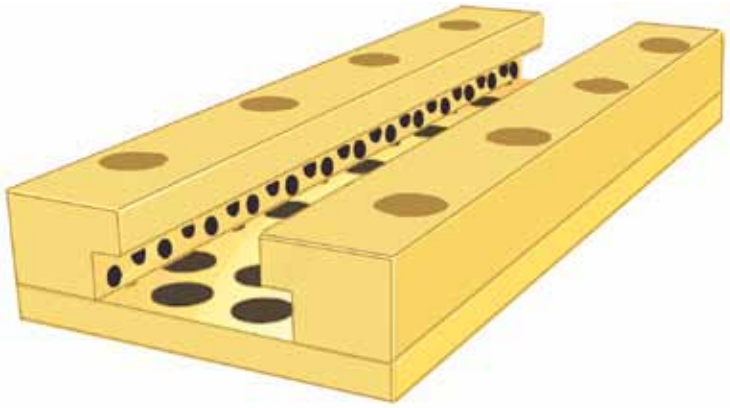
Bronze with
Self-Lubricating
Graphite Plugs



Precision machined sliding surfaces

Unless noted, all dimensions ±0.25mm

L-GIBS – METRIC – SOLID BRONZE – SELF-LUBRICATING															
PART NUMBER	L	W	A GH	B GW	C	D	E	F	G	H	J	K	QUANTITY		M & S Size
		+0.00 –0.13	+0.025	–0.000	+0.00 –0.13			±0.127					M	S	
L15LG100G	100	32	15	10	30	22	11	20	-	-	-	60	2	-	M10
L15LG150G	150								-	55	-	55	3	-	
L15LG200G	200								55	-	50	55	4	-	
L15LG250G	250								70	-	70	70	4	-	
L25LG200G	200	50	25	22	45	28	14		55	-	50	55	-	4	
L25LG250G	250								70	-	70	70	-	4	
L25LG300G	300								65	65	65	65	-	5	
L25LG350G	350								80	75	75	80	-	5	



Lamina off-the-shelf L-Gibs and Wear Strips can be put together to make your own custom assemblies..

The variety of sizes offered make custom gib and slide assembly construction fast and easy.

During normal sliding action, graphite produces a lubricating film across the entire wear surface.

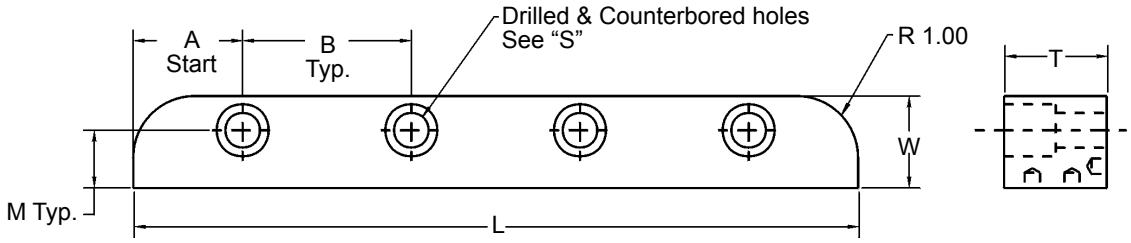
It is recommended that a mist of light oil be wiped on sliding surface prior to assembly.

One piece construction with pre-drilled and counterbored screw holes, machined complete.

Square Gibs

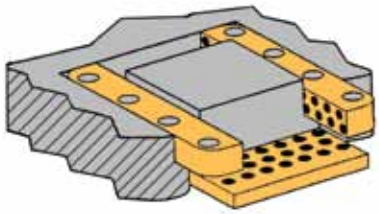


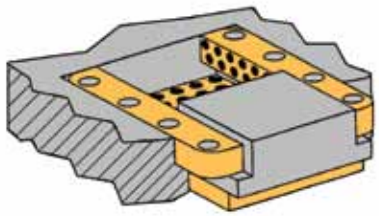
Bronze with
Self-Lubricating
Graphite Plugs



Unless noted, all Dimensions ±.010

SQUARE GIBS – BRONZE – SELF-LUBRICATING								
PART NUMBER	L	W	T	A	B	M	S	
		+.000 –.005	+.000 –.005		±.005	±.005	No. Holes	Size
L6SG32G	4.00	1.00	.75	1.000	2.000	.625	2	3/8″
L6SG40G	5.00				1.500		3	
L6SG48G	6.00				2.000		3	
L6SG64G	8.00				2.000		4	
L6SG80G	10.00				2.000		5	
L6SG96G	12.00				2.000		6	
L6SG96NG	12.00			NO MOUNTING HOLES				
L8SG32G	4.00	1.25	1.00	1.000	2.000	.875	2	3/8″
L8SG40G	5.00				1.500		3	
L8SG48G	6.00				2.000		3	
L8SG64G	8.00				2.000		4	
L8SG80G	10.00				2.000		5	
L8SG96G	12.00				2.000		6	
L8SG96NG	12.00			NO MOUNTING HOLES				
L12SG64G	8.00	2.00	1.50	1.500	2.500	1.250	3	5/8″
L12SG80G	10.00			1.250	2.500		4	
L12SG96G	12.00			1.500	3.000		4	
L12SG96NG	12.00			NO MOUNTING HOLES				





Create custom gib assemblies using Lamina component parts.

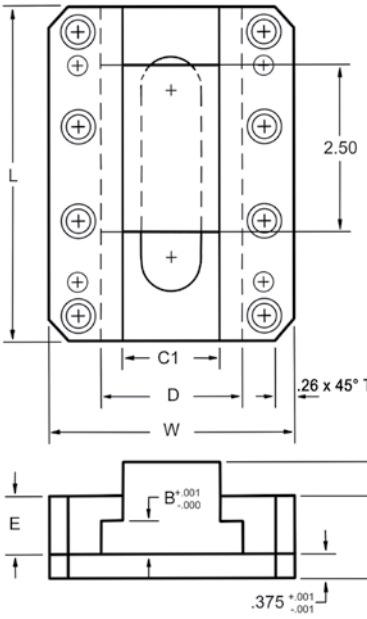
Lamina™

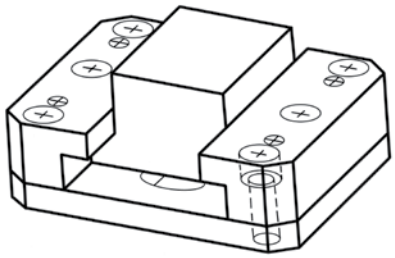
Gib Assemblies – Inch

- Plastics Injection Molds
- Special Machines
- Special Slide Applications



Bronze with
Self-Lubricating
Graphite Plugs





PRODUCT FEATURES

The gib assembly includes four components: two L-gibs, a base plate, and an optional T-slide. The L-gibs are provided with screw holes and are spot drilled for dowels. Also, the base plate includes thru holes to allow for easy assembly and fit into a pre-machined pocket. The gib assembly is a proven performer in a wide variety of applications, including lifter slides in molds.

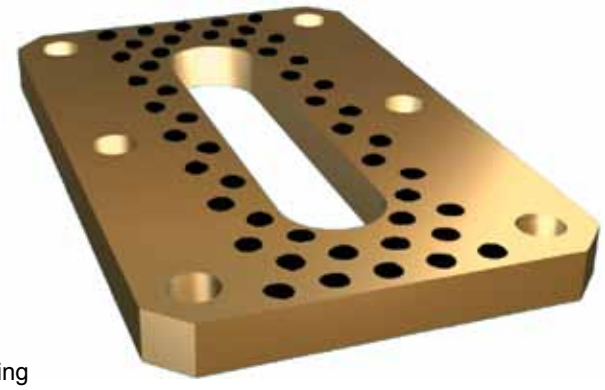
NOTE:

♦ Refer to page 11 for T-Slide dimensional information.

GIB ASSEMBLIES – INCH

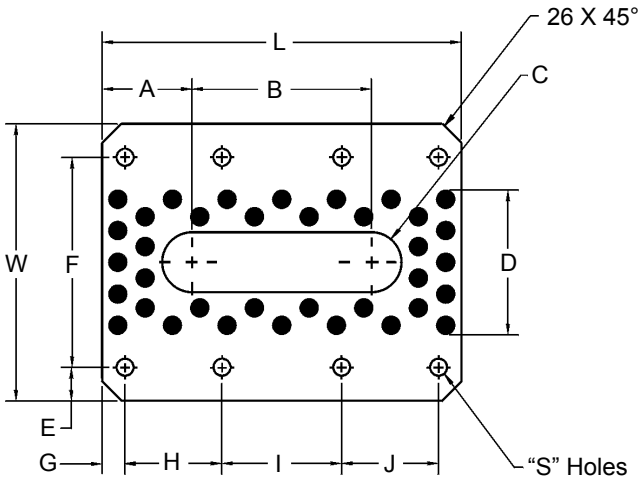
ASSY. WITH T-SLIDE PART NUMBER	ASSY. WITHOUT T-SLIDE PART NUMBER	L	W +.000 –.005	A	B +.001 –.000	C1	D	E	H	Socket Head Cap Screws		Locating Dowel Pins	
										No. Holes	Size	No. Holes	Size
L2GA16GTS	L2GA16G	2.00	2.620	1.12	.312	1.12	1.50	.75	1.63	4	1/4-20x1-1/4	4	1/4 x 1-1/2
L2GA24GTS	L2GA24G	3.00								4			
L2GA32GTS	L2GA32G	4.00								6			
L3GA24GTS	L3GA24G	3.00	3.120	1.12	.375	1.12	1.62	.75	1.62	4	5/16-18x1-1/2	4	1/4 x1-1/2
L3GA32GTS	L3GA32G	4.00								6			
L3GA40GTS	L3GA40G	5.00								6			
L4GA24GTS	L4GA24G	3.00	4.120	1.25	.500	1.62	2.37	.88	1.75	4	5/16-18x1-1/2	4	1/4 x1-1/2
L4GA32GTS	L4GA32G	4.00								6			
L4GA40GTS	L4GA40G	5.00								6			
L4GA48GTS	L4GA48G	6.00								8			
L6GA32GTS	L6GA32G	4.00	4.620	1.62	.750	1.62	2.62	1.25	2.13	6	5/16-18x1-3/4	4	1/4 x1-3/4
L6GA40GTS	L6GA40G	5.00								6			
L6GA48GTS	L6GA48G	6.00								8			

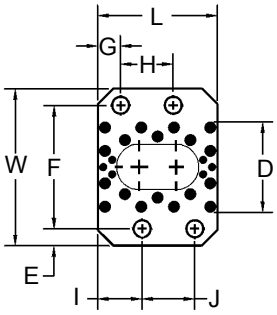
Base Plates for Gib Assemblies



Bronze with
Self-Lubricating
Graphite Plugs

NOTE:
- See page 9 for complete assembly.





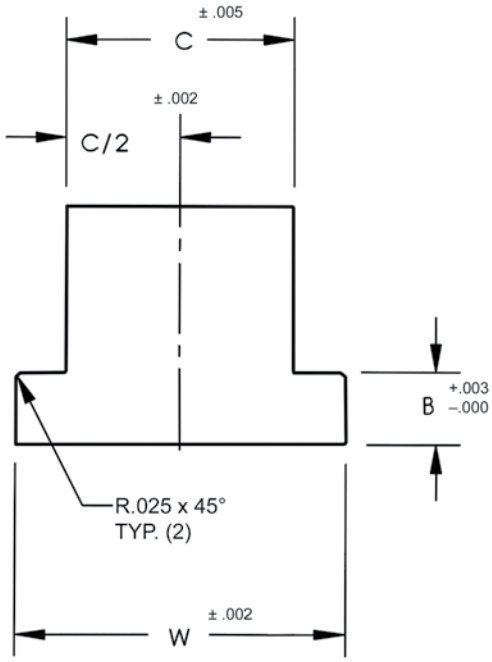
L2GAB16G
Hole Locations

Thickness = .375″

Unless noted, all dimensions ±.010

GIB ASSEMBLY BASE PLATES														
PART NUMBER	L	W +.000 –.005	A	B	C	D	E	F	G	H	I	J	S	
							±.005						NO.	SIZE
L2GAB16G	2.00	2.620	.69	.62	.38	1.50	.280	2.060	.380	.880	.740	.880	4	1/4″
L2GAB24G	3.00		.88	1.24						-	-	2.240	4	
L2GAB32G	4.00		1.00	2.00						1.620	-	1.620	6	
L3GAB24G	3.00	3.120	.88	1.24	.38	1.68	.370	2.380	.380	-	-	2.240	4	5/16″
L3GAB32G	4.00		1.00	2.00						1.620	-	1.620	6	
L3GAB40G	5.00		1.12	2.76						2.120	-	2.120	6	
L4GAB24G	3.00	4.120	1.00	1.00	.50	2.25	.495	3.130	.380	-	-	2.240	4	5/16″
L4GAB32G	4.00		1.12	1.76						1.620	-	1.620	6	
L4GAB40G	5.00		1.25	2.50						2.120	-	2.120	6	
L4GAB48G	6.00		1.50	3.00						1.620	2.000	1.620	8	
L6GAB32G	4.00	4.620	1.12	1.76	.50	2.41	.560	3.500	.380	1.620	-	1.620	6	5/16″
L6GAB40G	5.00		1.25	2.50						2.120	-	2.120	6	
L6GAB48G	6.00		1.50	3.00						1.620	2.000	1.620	8	

Optional T-Slide for Gib Assemblies – Inch

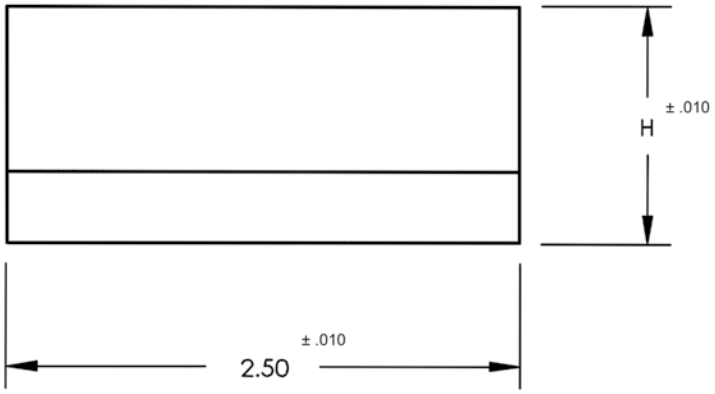


High quality carburized steel



NOTE:

- See page 9 for complete assembly.



PRODUCT FEATURES

The inch-sized T-Slide is often used as a Lifter Slide and can be purchased separately or as a part of the gib assembly.

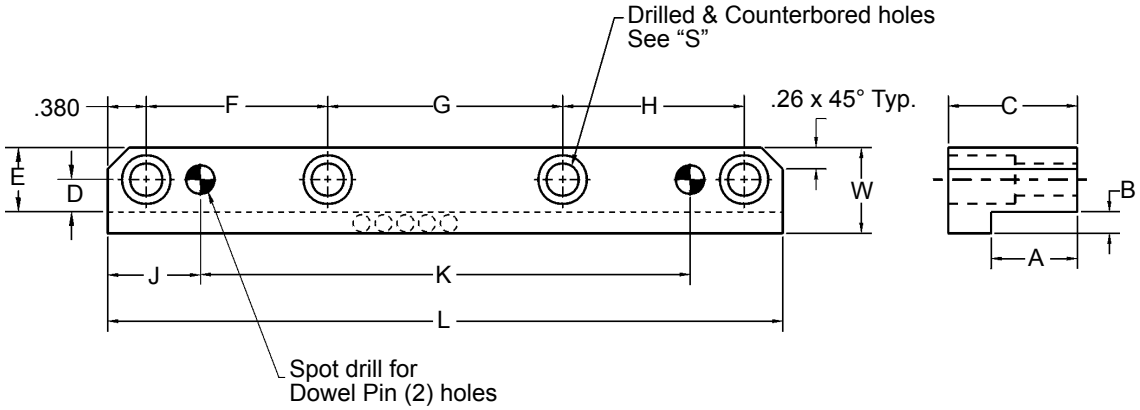
T-SLIDE FOR GIB ASSEMBLIES – INCH					
Part Number	Width W (in)	B (in)	C (in)	C/2 (in)	Height H (in)
2GASLIDE	1.493	0.308	1.110	0.555	1.25
3GASLIDE	1.617	0.371	1.110	0.555	1.25
4GASLIDE	2.367	0.496	1.610	0.805	1.38
6GASLIDE	2.617	0.746	1.610	0.805	1.75

L-Gibs for Gib Assemblies



Bronze with
Self-Lubricating
Graphite Plugs

NOTE:
- ♦ See page 9 for complete assembly.

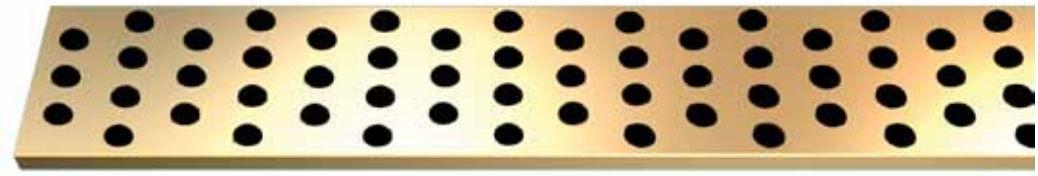


Precision machined sliding surface

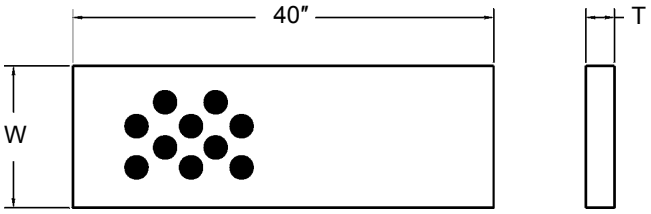
Unless noted, all dimensions ±.010

GIB ASSEMBLY L-GIBS														
PART NUMBER	L	W +.000 –.005	A GH +.001 –.000	B GW +.001 –.000	C +.000 –.005	D	E	F ±.005	G ±.005	H ±.005	J	K	S NO.	S SIZE
L2GAL16G	2.00	.75	.312	.188	.75	.280	.56	-	.880	-	.79	.95	2	1/4″
L2GAL24G	3.00							-	-	2.240	.88	1.24	2	
L2GAL32G	4.00							1.620	-	1.620	.88	2.24	3	
L3GAL24G	3.00	1.00	.375	.250	.75	.380	.75	-	-	2.240	.88	1.24	2	5/16″
L3GAL32G	4.00							1.620		1.620		2.24	3	
L3GAL40G	5.00							2.120		2.120		3.24	3	
L4GAL24G	3.00	1.25	.500	.375	.88	.380	.88	-	-	2.240	.88	1.24	2	
L4GAL32G	4.00							1.620	-	1.620		2.24	3	
L4GAL40G	5.00							2.120	-	2.120		3.24	3	
L4GAL48G	6.00							1.620	2.000	1.620		4.24	4	
L6GAL32G	4.00	1.50	.750	.500	1.25	.440	1.00	1.620	-	1.620	.88	2.24	3	
L6GAL40G	5.00							2.120	-	2.120		3.24	3	
L6GAL48G	6.00							1.620	2.000	1.620		4.24	4	

Wear Strip Ways – Solid Bronze – Graphite Plugs



Bronze with
Self-Lubricating
Graphite Plugs



Call Lamina Customer Service for hole pattern.

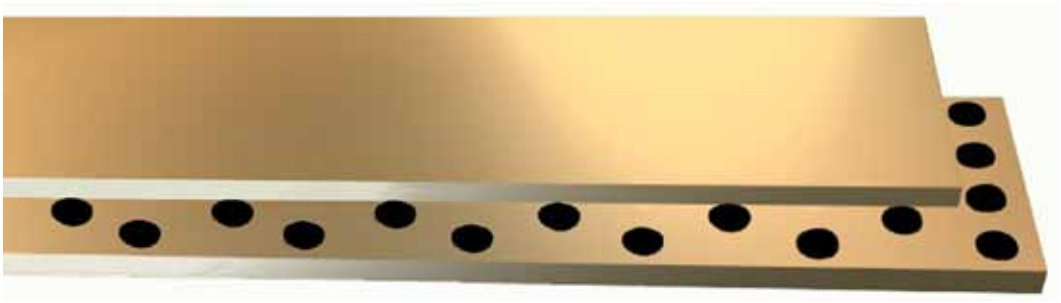
WEAR STRIP WAYS			
PART NUMBER	T ±.001	W	PLUG DIA.
L312320NG	0.375	1.50	5/16
L316320NG		2.00	3/8
L320320NG		2.50	3/8
L324320NG		3.00	1/2
L412320NG	0.500	1.50	5/16
L416320NG		2.00	3/8
L420320NG		2.50	3/8
L424320NG		3.00	1/2
L432320NG		4.00	1/2
L516320NG	0.625	2.00	3/8
L524320NG		3.00	1/2
L532320NG		4.00	1/2
L616320NG	0.750	2.00	3/8
L624320NG		3.00	1/2
L632320NG		4.00	1/2
L640320NG		5.00	1/2

Wear Strips – Bronze Plated

Lamina bronze-plated wear strips are offered cut to your exact length or in 48″ lengths.

WEAR STRIPS – INCH – BRONZE-PLATED											
BRONZE PLATED		T	W	BRONZE PLATED		T	W	BRONZE PLATED		T	W
PART NUMBER by the inch	PART NUMBER 48″ length	Thick-ness +.000 –.002	Width +.000 –.080	PART NUMBER by the inch	PART NUMBER 48″ length	Thick-ness +.000 –.002	Width +.000 –.080	PART NUMBER by the inch	PART NUMBER 48″ length	Thick-ness +.000 –.002	Width +.000 –.080
FP-1/8-x-1	PPF013100	1/8″	1	FP-5/16-x-1	PPF031100	5/16″	1	FP-5/8-x-1	PPF063100	5/8″	1
FP-1/8-x-1-1/4	PPF013125		1-1/4	FP-5/16-x-1-1/4	PPF031125		1-1/4	FP-5/8-x-1-1/4	PPF063125		1-1/4
FP-1/8-x-1-1/2	PPF013150		1-1/2	FP-5/16-x-1-1/2	PPF031150		1-1/2	FP-5/8-x-1-1/2	PPF063150		1-1/2
FP-1/8-x-1-3/4	PPF013175		1-3/4	FP-5/16-x-1-3/4	PPF031175		1-3/4	FP-5/8-x-1-3/4	PPF063175		1-3/4
FP-1/8-x-2	PPF013200		2	FP-5/16-x-2	PPF031200		2	FP-5/8-x-2	PPF063200		2
FP-1/8-x-2-1/2	PPF013250		2-1/2	FP-5/16-x-2-1/2	PPF031250		2-1/2	FP-5/8-x-2-1/2	PPF063250		2-1/2
FP-1/8-x-3	PPF013300		3	FP-5/16-x-3	PPF031300		3	FP-5/8-x-3	PPF063300		3
FP-1/8-x-3-1/2	PPF013350		3-1/2	FP-5/16-x-3-1/2	PPF031350		3-1/2	FP-5/8-x-3-1/2	PPF063350		3-1/2
FP-1/8-x-4	PPF013400		4	FP-5/16-x-4	PPF031400		4	FP-5/8-x-4	PPF063400		4
FP-1/8-x-4-1/2	PPF013450		4-1/2	FP-5/16-x-4-1/2	PPF031450		4-1/2	FP-5/8-x-4-1/2	PPF063450		4-1/2
FP-1/8-x-5	PPF013500		5	FP-5/16-x-5	PPF031500		5	FP-5/8-x-5	PPF063500		5
FP-1/8-x-6	PPF013600		6	FP-5/16-x-6	PPF031600		6	FP-5/8-x-6	PPF063600		6
FP-3/16-x-1	PPF019100	3/16″	1	FP-5/16-x-8	PPF031800		8	FP-5/8-x-8	PPF063800		8
FP-3/16-x-1-1/4	PPF019125		1-1/4	FP-5/16-x-10	PPF0311000		10	FP-5/8-x-10	PPF0631000		10
FP-3/16-x-1-1/2	PPF019150		1-1/2	FP-5/16-x-12	PPF0311200		12	FP-5/8-x-12	PPF0631200		12
FP-3/16-x-1-3/4	PPF019175		1-3/4	FP-3/8-x-1	PPF038100	3/8″	1	FP-3/4-x-1	PPF075100	3/4″	1
FP-3/16-x-2	PPF019200		2	FP-3/8-x-1-1/4	PPF038125		1-1/4	FP-3/4-x-1-1/4	PPF075125		1-1/4
FP-3/16-x-2-1/2	PPF019250		2-1/2	FP-3/8-x-1-1/2	PPF038150		1-1/2	FP-3/4-x-1-1/2	PPF075150		1-1/2
FP-3/16-x-3	PPF019300		3	FP-3/8-x-1-3/4	PPF038175		1-3/4	FP-3/4-x-1-3/4	PPF075175		1-3/4
FP-3/16-x-3-1/2	PPF019350		3-1/2	FP-3/8-x-2	PPF038200		2	FP-3/4-x-2	PPF075200		2
FP-3/16-x-4	PPF019400		4	FP-3/8-x-2-1/2	PPF038250		2-1/2	FP-3/4-x-2-1/2	PPF075250		2-1/2
FP-3/16-x-4-1/2	PPF019450		4-1/2	FP-3/8-x-3	PPF038300		3	FP-3/4-x-3	PPF075300		3
FP-3/16-x-5	PPF019500		5	FP-3/8-x-3-1/2	PPF038350		3-1/2	FP-3/4-x-3-1/2	PPF075350		3-1/2
FP-3/16-x-6	PPF019600		6	FP-3/8-x-4	PPF038400		4	FP-3/4-x-4	PPF075400		4
FP-3/16-x-8	PPF019800		8	FP-3/8-x-4-1/2	PPF038450		4-1/2	FP-3/4-x-4-1/2	PPF075450		4-1/2
FP-1/4-x-1	PPF025100	1/4″	1	FP-3/8-x-5	PPF038500		5	FP-3/4-x-5	PPF075500		5
FP-1/4-x-1-1/4	PPF025125		1-1/4	FP-3/8-x-6	PPF038600		6	FP-3/4-x-6	PPF075600		6
FP-1/4-x-1-1/2	PPF025150		1-1/2	FP-3/8-x-8	PPF038800		8	FP-3/4-x-8	PPF075800		8
FP-1/4-x-1-3/4	PPF025175		1-3/4	FP-3/8-x-10	PPF0381000		10	FP-3/4-x-10	PPF0751000		10
FP-1/4-x-2	PPF025200		2	FP-3/8-x-12	PPF0381200		12	FP-3/4-x-12	PPF0751200		12
FP-1/4-x-2-1/2	PPF025250		2-1/2	FP-1/2-x-1	PPF050100	1/2″	1	FP-1-x-1	PPF100100	1″	1
FP-1/4-x-3	PPF025300		3	FP-1/2-x-1-1/4	PPF050125		1-1/4	FP-1-x-1-1/4	PPF100125		1-1/4
FP-1/4-x-3-1/2	PPF025350		3-1/2	FP-1/2-x-1-1/2	PPF050150		1-1/2	FP-1-x-1-1/2	PPF100150		1-1/2
FP-1/4-x-4	PPF025400		4	FP-1/2-x-1-3/4	PPF050175		1-3/4	FP-1-x-1-3/4	PPF100175		1-3/4
FP-1/4-x-4-1/2	PPF025450		4-1/2	FP-1/2-x-2	PPF050200		2	FP-1-x-2	PPF100200		2
FP-1/4-x-5	PPF025500		5	FP-1/2-x-2-1/2	PPF050250		2-1/2	FP-1-x-2-1/2	PPF100250		2-1/2
FP-1/4-x-6	PPF025600		6	FP-1/2-x-3	PPF050300		3	FP-1-x-3	PPF100300		3
FP-1/4-x-8	PPF025800		8	FP-1/2-x-3-1/2	PPF050350		3-1/2	FP-1-x-3-1/2	PPF100350		3-1/2
FP-1/4-x-10	PPF0251000		10	FP-1/2-x-4	PPF050400		4	FP-1-x-4	PPF100400		4
FP-1/4-x-12	PPF0251200		12	FP-1/2-x-4-1/2	PPF050450		4-1/2	FP-1-x-4-1/2	PPF100450		4-1/2
Create custom assemblies using standard Lamina component parts.				FP-1/2-x-5	PPF050500		5	FP-1-x-5	PPF100500		5
				FP-1/2-x-6	PPF050600		6	FP-1-x-6	PPF100600		6
				FP-1/2-x-8	PPF050800		8	FP-1-x-8	PPF100800		8
				FP-1/2-x-10	PPF0501000		10	FP-1-x-10	PPF1001000		10
				FP-1/2-x-12	PPF0501200		12	FP-1-x-12	PPF1001200		12

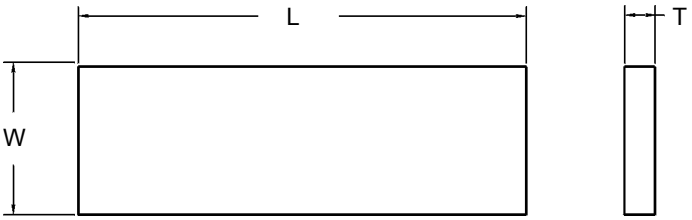
Wear Strips – Bronze-Plated



Standard wear strips are plated on one side only.

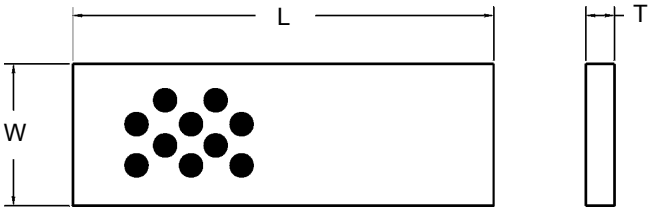
Up to four sides can be plated. Please call for quotation.

PLAIN



Maximum length 48″

SELF-LUBRICATING



Maximum length 48″

In order to be flat this material must be fastened to a flat surface.

- **Parallel .001 within 4′**
- **Thickness of bronze: .008 to .010**
- **Milled edges**

Machining may cause distortion which can result in the loss of flatness of the part. Once altered, Lamina will not replace wear strips. We can provide specials at a reasonable price. Send your prints to sales@lamina.com

Wear Strips – Bronze-Plated & Self-Lubricating

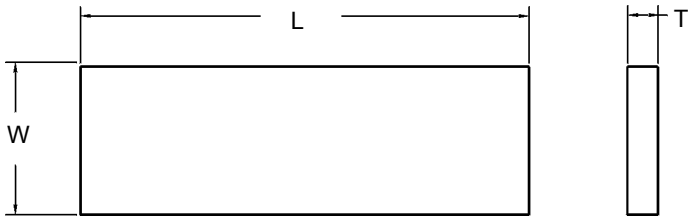
WEAR STRIPS – INCH – BRONZE-PLATED – SELF-LUBRICATING

PART NUMBER		T	W	PART NUMBER		T	W	PART NUMBER		T	W
By the inch	48″ length	Thick-ness +.000 –.002	Width +.000 –.080	By the inch	48″ length	Thick-ness +.000 –.002	Width +.000 –.080	By the inch	48″ length	Thick-ness +.000 –.002	Width +.000 –.080
SLFP025-100	SLF025100	1/4″	1	SLFP050-100	SLF050100	1/2″	1	SLFP100-100	SLF100100	1″	1
SLFP025-150	SLF025150		1-1/2	SLFP050-150	SLF050150		1-1/2	SLFP100-150	SLF100150		1-1/2
SLFP025-200	SLF025200		2	SLFP050-200	SLF050200		2	SLFP100-200	SLF100200		2
SLFP025-250	SLF025250		2-1/2	SLFP050-250	SLF050250		2-1/2	SLFP100-250	SLF100250		2-1/2
SLFP025-300	SLF025300		3	SLFP050-300	SLF050300		3	SLFP100-300	SLF100300		3
SLFP025-400	SLF025400		4	SLFP050-400	SLF050400		4	SLFP100-400	SLF100400		4
SLFP025-500	SLF025500		5	SLFP050-500	SLF050500		5	SLFP100-500	SLF100500		5
SLFP025-600	SLF025600		6	SLFP050-600	SLF050600		6	SLFP100-600	SLF100600		6
SLFP038-100	SLF038100	3/8″	1	SLFP075-100	SLF075100	3/4″	1	**These products are in stock at Lamina and at Lamina stocking distributors throughout the world.**			
SLFP038-150	SLF038150		1-1/2	SLFP075-150	SLF075150		1-1/2				
SLFP038-200	SLF038200		2	SLFP075-200	SLF075200		2				
SLFP038-250	SLF038250		2-1/2	SLFP075-250	SLF075250		2-1/2				
SLFP038-300	SLF038300		3	SLFP075-300	SLF075300		3				
SLFP038-400	SLF038400		4	SLFP075-400	SLF075400		4				
SLFP038-500	SLF038500		5	SLF0075-500	SLF075500		5				
SLFL038-600	SLF038600		6	SLFP075-600	SLF075600		6				

Lamina bronze-plated self-lubricating wear strips are offered cut to your length or in 48″ lengths. We suggest cutting these strips in 1/2″ increments to avoid cutting through the plugs.

Wear Strips – Bronze-Plated – FP Metric





Maximum length 1219 mm

Standard wear strips are plated on one side only.

Up to four sides can be plated, please call for a quotation.

- **In order to be flat this material must be fastened to a flat surface.**
- **Parallel 0.025 within 1219mm**
- **Thickness of bronze: 0.20 to 0.25**
- **Milled edges**

Machining may cause distortion which can result in the loss of flatness of the part. Once altered, Lamina will not replace wear strips. We can provide specials at a reasonable price. Send your prints to sales@lamina.com

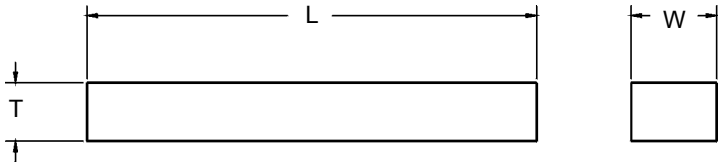
WEAR STRIPS – METRIC – BRONZE-PLATED

BRONZE PLATED		T Thick-ness	W Width	BRONZE PLATED		T Thick-ness	W Width	BRONZE PLATED		T Thick-ness	W Width
PART NUMBER by 25mm	PART NUMBER 1219mm length	+0.0 –0.051	+0.000 –2.032	PART NUMBER by 25mm	PART NUMBER 1219mm length	+0.0 –0.051	+0.000 –2.032	PART NUMBER by 25mm	PART NUMBER 1219mm length	+0.0 –0.051	+0.000 –2.032
FP-4MM-X-25	MFP04025	4mm	25mm	FP-10MM-X-25	MFP10025	10mm	25mm	FP-20MM-X-30	MFP20030	20mm	30mm
FP-4MM-X-30	MFP04030		30mm	FP-10MM-X-30	MFP10030		30mm	FP-20MM-X-50	MFP20050		50mm
FP-4MM-X-35	FMP04035		35mm	FP-10MM-X-35	MFP10035		35mm	FP-20MM-X-60	MFP20060		60mm
FP-4MM-X-40	MFP04040		40mm	FP-10MM-X-40	MFP10040		40mm	FP-20MM-X-75	MFP20075		75mm
FP-4MM-X-60	MFP04060		60mm	FP-10MM-X-50	MFP10050		50mm	FP-20MM-X-100	MFP20100		100mm
FP-4MM-X-100	MFP04100		100mm	FP-10MM-X-60	MFP10060		60mm	FP-25MM-X-25	MFP25025	25mm	25mm
FP-5MM-X-25	MFP05025	5mm	25mm	FP-10MM-X-75	MFP10075		75mm	FP-25MM-X-100	MFP25100		100mm
FP-5MM-X-30	MFP05030		30mm	FP-10MM-X-100	MFP10100		100mm				
FP-5MM-X-40	MFP05040		40mm	FP-10MM-X-125	MFP10125		125mm				
FP-5MM-X-75	MFP05075		75mm	FP-10MM-X-150	MFP10150		150mm				
FP-5MM-X-100	MFP05100		100mm	FP-12MM-X-50	MFP12050	12mm	50mm				
FP-6MM-X-25	MFP06025	6mm	25mm	FP-12MM-X-75	MFP12075		75mm				
FP-6MM-X-30	MFP06030		30mm	FP-12MM-X-100	MFP12100		100mm				
FP-6MM-X-35	MFP06035		35mm	FP-12MM-X-125	MFP12125		125mm				
FP-6MM-X-40	MFP06040		40mm	FP-12MM-X-150	MFP12150		150mm				
FP-6MM-X-50	MFP06050		50mm	FP-12MM-X-200	MFP12200		200mm				
FP-6MM-X-60	MFP06060		60mm	FP-15MM-X-25	MFP15025	15mm	25mm				
FP-6MM-X-75	MFP06075		75mm	FP-15MM-X-40	MFP15040		40mm				
FP-8MM-X-25	MFP08025	8mm	25mm	FP-15MM-X-60	MFP15060		60mm				
FP-8MM-X-60	MFP08060		60mm	FP-15MM-X-75	MFP15075		75mm				
FP-8MM-X-75	MFP08075		75mm	FP-15MM-X-100	MFP15100		100mm				
FP-8MM-X-100	MPF08100		100mm	FP-15MM-X-125	MFP15125		125mm				
FP-8MM-X-125	MFP08125		125mm	FP-15MM-X-150	MFP15150		150mm				

Lamina is the only manufacturer of bronze-plated wear strips.

Side Strip – Bronze-Plated – SP Inch & Metric





NOTE:

- Maximum length is 48″ (1219mm).

SIDE STRIP – METRIC			
BRONZE-PLATED		T Thickness –0.45 –0.50	W Width +0.0 –0.12
PART NUMBER by 25mm	PART NUMBER 1219mm length		
SP-15MM-X-25MM	MSP15025	15	25
SP-25MM-X-37MM	MSP25037	25	37

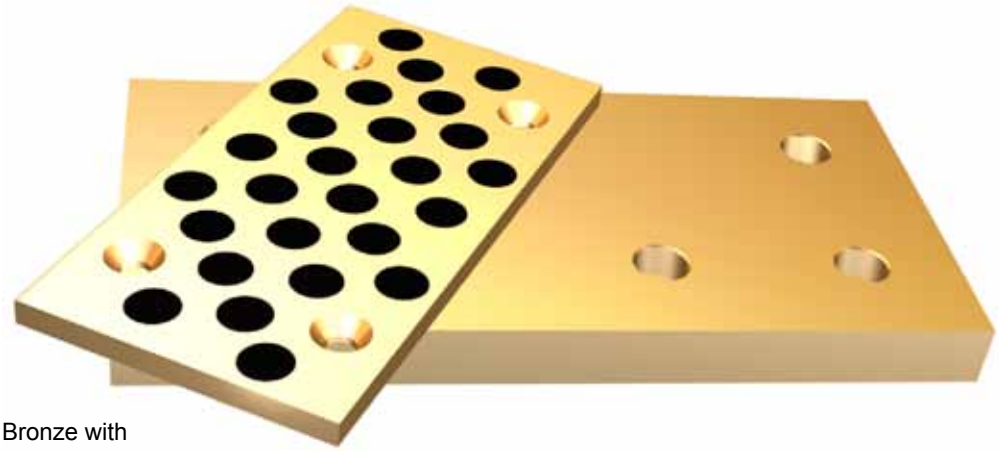
SIDE STRIP –INCH			
BRONZE-PLATED		T THICK-NESS –.018 –.020	W WIDTH –.000 –.005
PART NUMBER by the Inch	PART NUMBER 48″ length		
SP-1/2-X-1	SPF050100	1/2	1
SP-1/2-X-1-1/2	SPF050150		1-1/2
SP-5/8-X-1	SPF063100	5/8	1
SP-5/8-X-2	SPF063200		2
SP-3/4-X-1	SPF075100	3/4	1
SP-3/4-X-1-1/2	SPF075150		1-1/2
SP-3/4-X-2	SPF075200		2
SP-1-X-1-1/2	SPF100150	1	1-1/2
SP-1-X-2	SPF100200		2
SP-1-X-3	SPF100300		3
SP-1-1/4-X-2	SPF125200	1-1/4	2
SP-1-1/4-X-3	SPF125300		3
SP-1-1/2-X-2	SPF150200	1-1/2	2
SP-1-1/2-X-2-1/2	SPF150250		2-1/2
SP-2-X-2	SPF200200	2	2

Wear Strips – Inch – Fixed-Length

No suffix for solid bronze
G or NG is necessary for ordering self-lubricating – See Page 19.

WEAR STRIPS – IMPERIAL															
PART NUMBERS (Thickness)				W	L	SCREW HOLES		HOLE LOCATION							
1/4″	3/8″	1/2″	5/8″	Width	Length	AMT	PAT	A	B	C	D	E	F	G	H
L28-32	L38-32	-	-	1.00	4.00	2	A	1.00	3.00	-	-	-	-	.50	-
L28-40	L38-40	-	-		5.00	3		.50	2.50	4.50	-	-	-		
L28-48	L38-48	-	-		6.00	3		1.00	3.00	5.00	-	-	-		
L28-64	L38-64	-	-		8.00	4		1.00	3.00	5.00	7.00	-	-		
L28-80	L38-80	-	-		10.00	5		1.00	3.00	5.00	7.00	9.00	-		
L28-96	L38-96	-	-		12.00	6		1.00	3.00	5.00	7.00	9.00	11.00		
-	-	L412-24	L512-24	1.50	3.00	2	A	.50	2.50	-	-	-	-	.75	-
L212-32	L312-32	L412-32	L512-32		4.00	2		1.00	3.00	-	-	-	-		
L212-40	L312-40	L412-40	L512-40		5.00	3		.50	2.50	4.50	-	-	-		
L212-48	L312-48	L412-48	L512-48		6.00	3		1.00	3.00	5.00	-	-	-		
L212-64	L312-64	L412-64	L512-64		8.00	4		1.00	3.00	5.00	7.00	-	-		
L212-80	L312-80	L412-80	L512-80		10.00	5		1.00	3.00	5.00	7.00	9.00	-		
L212-96	L312-96	L412-96	L512-96		12.00	6		1.00	3.00	5.00	7.00	9.00	11.00		
-	-	L416-24	L516-24	2.00	3.00	2	A	.50	2.50	-	-	-	-	1.00	-
L216-32	L316-32	L416-32	L516-32		4.00	2		1.00	3.00	-	-	-	-		
L216-40	L316-40	L416-40	L516-40		5.00	3		.50	2.50	4.50	-	-	-		
L216-48	L316-48	L416-48	L516-48		6.00	3		1.00	3.00	5.00	-	-	-		
L216-64	L316-64	L416-64	L516-64		8.00	4		1.00	3.00	5.00	7.00	-	-		
L216-80	L316-80	L416-80	L516-80		10.00	5		1.00	3.00	5.00	7.00	9.00	-		
L216-96	L316-96	L416-96	L516-96		12.00	6		1.00	3.00	5.00	7.00	9.00	11.00		
-	-	L420-24	L520-24	2.50	3.00	2	A	.50	2.50	-	-	-	-	1.25	-
L220-32	L320-32	L420-32	L520-32		4.00	2		1.00	3.00	-	-	-	-		
L220-40	L320-40	L420-40	L520-40		5.00	3		.50	2.50	4.50	-	-	-		
L220-48	L320-48	L420-48	L520-48		6.00	3		1.00	3.00	5.00	-	-	-		
L220-64	L320-64	L420-64	L520-64		8.00	4		1.00	3.00	5.00	7.00	-	-		
L220-80	L320-80	L420-80	L520-80		10.00	5		1.00	3.00	5.00	7.00	9.00	-		
L220-96	L320-96	L420-96	L520-96		12.00	6		1.00	3.00	5.00	7.00	9.00	11.00		
-	-	L424-24	L524-24	3.00	3.00	4	B	.50	2.50	-	-	-	-	0.50	2.50
L224-32	L324-32	L424-32	L524-32		4.00	4		1.00	3.00	-	-	-	-		
L224-40	L324-40	L424-40	L524-40		5.00	4		.50	4.50	-	-	-	-		
L224-48	L324-48	L424-48	L524-48		6.00	4		1.00	5.00	-	-	-	-		
L224-64	L324-64	L424-64	L524-64		8.00	6	C	1.00	3.00	5.00	-	-	7.00		
L224-80	L324-80	L424-80	L524-80		10.00	6	B	1.00	5.00	9.00	-	-	-		
L224-96	L324-96	L424-96	L524-96		12.00	8	C	1.00	3.00	5.00	7.00	9.00	11.00		
L232-32	L332-32	L432-32	L532-32	4.00	4.00	4	B	1.00	3.00	-	-	-	-	1.00	3.00
L232-40	L332-40	L432-40	L532-40		5.00	4		.50	4.50	-	-	-	-		
L232-48	L332-48	L432-48	L532-48		6.00	4		1.00	5.00	-	-	-	-		
L232-64	L332-64	L432-64	L532-64		8.00	6	C	1.00	3.00	5.00	-	-	7.00		
L232-80	L332-80	L432-80	L532-80		10.00	6	B	1.00	5.00	9.00	-	-	-		
L232-96	L332-96	L432-96	L532-96		12.00	8	C	1.00	3.00	5.00	7.00	9.00	11.00		

Wear Strips – Inch – Fixed-Length



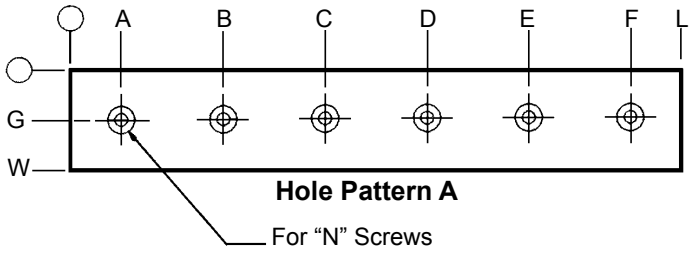
Bronze with
Self-Lubricating
Graphite Plugs

- **Bronze with self-lubricating graphite plugs for bilateral wear**
- **Ready to use**
- **Solid Bronze available upon request**

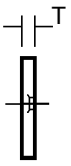
When ordering please indicate preference at the end of the part number:

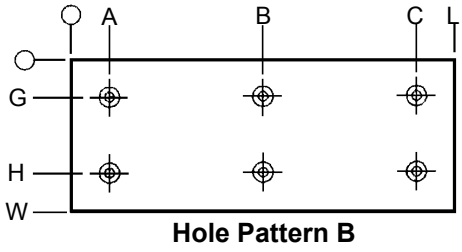
G	Drilled with Graphite	Example: L212-32G
NG	Undrilled with graphite	Example: L212-32NG

Mounting hole patterns for Wear Strips



Hole Pattern A

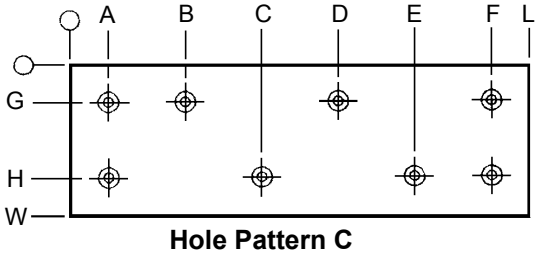




Hole Pattern B

Material: Bronze
All Dimensions ±.010 unless noted

RECOMMENDED MOUNTING SCREW SIZES	
THICKNESS	N
1/4	1/4″ Flat head screw
3/8	1/4″ Socket head cap screw
1/2	5/16″ Socket head cap screw
5/8	5/16″ Socket head cap screw

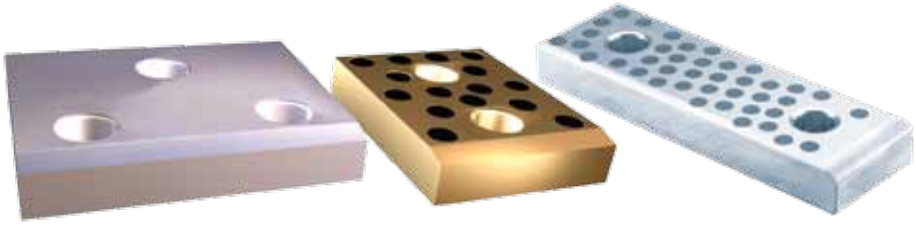


Hole Pattern C

NOTE:

- Width and length of wearstrip determines hole pattern, regardless of thickness.

Wear Plates – NAAMS Style W05, W06 & W07/VDI 3357 – Metric



Steel | **Bronze - Self Lubricating** | **Steel - Self-Lubricating**

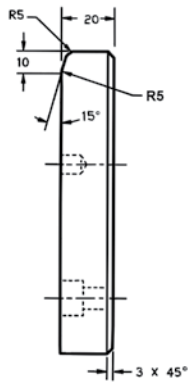
PRODUCT FEATURES

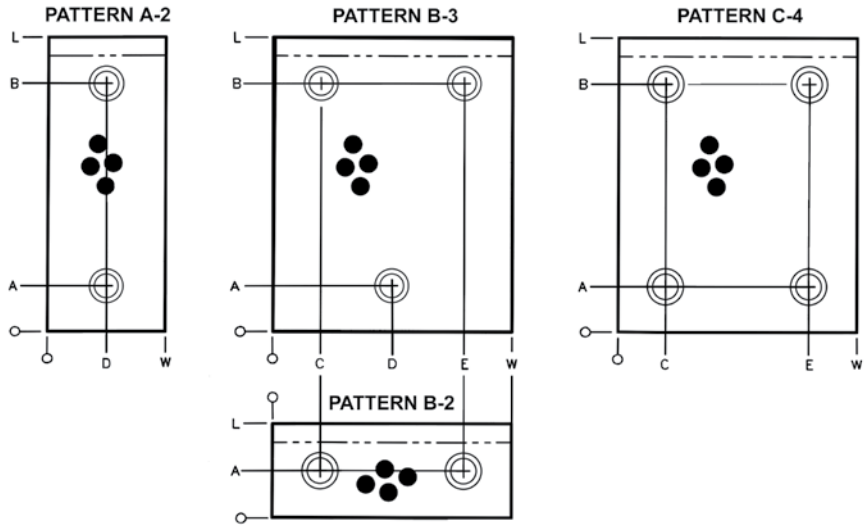
These wear plates are 20mm thick and available in bronze material with graphite, steel with graphite and plain steel. They are also available in the GM sizes in cast iron with graphite.

VDI/NAAMS STANDARDS

Width W (mm)	Length L (mm)	Bronze w/Graphite Plugs	Steel w/Graphite Plugs	Steel Plain	Mounting Hole Pattern & Location						
					Hole	Hole	A	B	C	D	E
		Part Number	Part Number	Part Number	Qty.	Pattern	(mm)	(mm)	(mm)	(mm)	(mm)
50	80	VDY050080	VDG050080	VDX050080	2	A-2	25	55	–	25	–
	100	VDY050100	VDG050100	VDX050100	2	A-2	25	75	–	25	–
	125	VDY050125	VDG050125	VDX050125	2	A-2	25	100	–	25	–
	160	VDY050160	VDG050160	VDX050160	2	A-2	25	135	–	25	–
	200	VDY050200	VDG050200	VDX050200	2	A-2	25	175	–	25	–
80	50	VDY080050	VDG080050	VDX080050	2	B-2	25	–	25	–	55
	80	VDY080080	VDG080080	VDX080080	2	A-2	25	55	–	40	–
	100	VDY080100	VDG080100	VDX080100	2	A-2	25	75	–	40	–
	125	VDY080125	VDG080125	VDX080125	2	A-2	25	100	–	40	–
	160	VDY080160	VDG080160	VDX080160	2	A-2	25	135	–	40	–
	200	VDY080200	VDG080200	VDX080200	2	A-2	25	175	–	40	–
	250	VDY080250	VDG080250	VDX080250	2	A-2	40	210	–	40	–
	315	VDY080315	VDG080315	VDX080315	2	A-2	40	275	–	40	–
100	50	VDY100050	VDG100050	VDX100050	2	B-2	25	–	25	–	75
	80	VDY100080	VDG100080	VDX100080	2	B-2	40	–	25	–	75
	100	VDY100100	VDG100100	VDX100100	2	A-2	25	75	–	50	–
	125	VDY100125	VDG100125	VDX100125	2	A-2	25	100	–	50	–
	160	VDY100160	VDG100160	VDX100160	2	A-2	25	135	–	50	–
	200	VDY100200	VDG100200	VDX100200	2	A-2	25	175	–	50	–
	250	VDY100250	VDG100250	VDX100250	2	A-2	40	210	–	50	–
	315	VDY100315	VDG100315	VDX100315	2	A-2	40	275	–	50	–
125	50	VDY125050	VDG125050	VDX125050	2	B-2	25	–	25	–	100
	80	VDY125080	VDG125080	VDX125080	2	B-2	40	–	25	–	
	100	VDY125100	VDG125100	VDX125100	3	B-3	25	75	25	62.5	
	125	VDY125125	VDG125125	VDX125125	3	B-3	25	100	25	62.5	
	160	VDY125160	VDG125160	VDX125160	3	B-3	25	135	25	62.5	
	200	VDY125200	VDG125200	VDX125200	3	B-3	25	175	25	62.5	
	250	VDY125250	VDG125250	VDX125250	3	B-3	40	210	25	62.5	
	315	VDY125315	VDG125315	VDX125315	3	B-3	40	275	25	62.5	
160	50	VDY160050	VDG160050	VDX160050	2	B-2	25	–	25	–	135
	80	VDY160080	VDG160080	VDX160080	2	B-2	40	–	25	–	
	100	VDY160100	VDG160100	VDX160100	3	B-3	25	75	25	80	
	125	VDY160125	VDG160125	VDX160125	3	B-3	25	100	25	80	
	160	VDY160160	VDG160160	VDX160160	3	B-3	25	135	25	80	
	200	VDY160200	VDG160200	VDX160200	3	B-3	25	175	25	80	
	250	VDY160250	VDG160250	VDX160250	4	C-4	40	210	25	–	
	315	VDY160315	VDG160315	VDX160315	4	C-4	40	275	25	–	

Wear Plates – NAAMS Series W05, W06 & W07/VDI 3357 – Metric





MATERIALS:

- Self-lubricating wear plates are manufactured from bronze or steel with graphite plugs. The plugs cover 20-25% of the wear surface.
- Wear plates are also available in plain steel.
- Wear plates are also available in the GM sizes in cast iron with graphite. See the cross reference sheet on pg. 23 for the GM part number.

TOLERANCES:

- Thickness ± 0.00/ -0.02 mm
- Mounting hole locations ± 0.2 mm
- Flatness ± 0.013 mm
- Unless otherwise noted ± 0.25 mm

MOUNTING SCREWS:

- M8 S.H.C.S. for part numbers VDY050080 and VDY080050.
- M12 S.H.C.S. for all other part numbers.

ADDITIONAL AUTOMOTIVE STANDARDS

Width W (mm)	Length L (mm)	Bronze w/Graphite Plugs Part Number	Steel w/Graphite Plugs Part Number	Steel Plain Part Number	Mounting Hole Pattern & Location						
					Hole Qty.	Hole Pattern	A (mm)	B (mm)	C (mm)	D (mm)	E (mm)
100	250	WDX13-80-03100250Y*	WDX13-80-03100250G*	WDX13-80-03100250X*	2	A-2	25	225	–	50	–
	250	90.20.65-100250Y*	–	90.20.65-100250X*	4	C-4	25	225	25	–	75
	315	WDX13-80-03100315Y*	WDX13-80-03100315G*	WDX13-80-03100315X*	2	A-2	25	290	–	50	–
125	200	90.20.65-125200*	–	90.20.65-125200*	4	C-4	25	175	25	–	100
	250	WDX13-80-03125250Y*	WDX13-80-03125250G*	WDX13-80-03125250X*	3	B-3	25	225	25	62.5	100
	250	90.20.65-125250Y*	–	90.20.65-125250X*	4	C-4	25	225	25	–	100
	315	WDX13-80-03125315Y*	WDX13-80-03125315G*	WDX13-80-03125315X*	3	B-3	25	290	25	62.5	100
160	200	90.20.65-160200Y*	–	90.20.65-160200G*	4	C-4	25	175	25	–	135
	250	WDX13-80-03160250Y*	WDX13-80-03160250G*	WDX13-80-03160250X*	3	B-3	25	225	25	80	135
	250	90.20.65-160250Y*	–	90.20.65-160250X*	4	C-4	25	225	25	–	135
	300	90.20.65-160300Y*	–	90.20.65-160300X*	4	C-4	25	275	25	–	135
200	200	90.20.65-200200Y*	–	90.20.65-200200X*	4	C-4	25	175	25	–	175
	250	90.20.65-200250Y*	–	90.20.65-200250X*	4	C-4	25	225	25	–	175
	300	90.20.65-200300Y*	–	90.20.65-200300X*	4	C-4	25	275	25	–	175

PLEASE NOTE:

- These wear plates have hole patterns that differ from our standard VDI/NAAMS wear plates. They must be ordered using the GM or Ford part number to receive the correct mounting hole locations.

Lamina™

Automotive Wear Plate Conversions

Dimensions		BRONZE WITH GRAPHITE PLUGS					STEEL WITH GRAPHITE PLUGS		
Width	Length	Anchor Lamina Part Number	NAAMS W07 Series Number	Chrysler Part Number	Ford Part Number Part Number	General Motors Part Number	Anchor Lamina Part Number	NAAMS W05 Series Number	Ford Part Number
50	80	VDY050080					VDG20050080		
50	100	VDY050100	W070510	19-290-3204	WDX138030501100Y	90.20.65-50100Y	VDG20050100	W050510	WDX138030501100G
50	125	VDY050125	W070512	19-290-3205	WDX138030501125Y	90.20.65-50125Y	VDG20050125	W050512	WDX138030501125G
50	160	VDY050160	W070516	19-290-3206	WDX138030501160Y	90.20.65-50160Y	VDG20050160	W050516	WDX138030501160G
50	200	VDY050200	W070520	19-290-3208	WDX138030501200Y	90.20.65-50200Y	VDG20050200	W050520	WDX138030501200G
80	50	VDY080050					VDG20080050		
80	80	VDY080080					VDG20080080		
80	100	VDY080100	W070810	19-290-3304	WDX138030801100Y	90.20.65-80100Y	VDG20080100	W050810	WDX138030801100G
80	125	VDY080125	W070812	19-290-3305	WDX138030801125Y	90.20.65-80125Y	VDG20080125	W050812	WDX138030801125G
80	160	VDY080160	W070816	19-290-3306	WDX138030801160Y	90.20.65-80160Y	VDG20080160	W050816	WDX138030801160G
80	200	VDY080200	W070820	19-290-3308	WDX138030802200Y	90.20.65-80200Y	VDG20080200	W050820	WDX138030802200G
80	250	VDY080250		19-290-3310			VDG20080250		
80	315	VDY080315		19-290-3312			VDG20080315		
100	50	VDY100050		19-290-3402		90.20.65-10050Y	VDG20100050		
100	80	VDY100080	W071080	19-290-3403	WDX138031000800Y	90.20.65-10080Y	VDG20100080	W051080	WDX138031000800G
100	100	VDY100100	W071010	19-290-3404	WDX138031001100Y	90.20.65-10010OY	VDG20100100	W051010	WDX138031001100G
100	125	VDY100125	W071012	19-290-3405	WDX138031001125Y	90.20.65-10012SY	VDG20100125	W051012	WDX138031001125G
100	160	VDY100160	W071016	19-290-3406	WDX138031001160Y	90.20.65-10016OY	VDG20100160	W051016	WDX138031001160G
100	200	VDY100200	W071020	19-290-3408	WDX138031002200Y	90.20.65-10020OY	VDG20100200	W051020	WDX138031002200G
100	250	VDY100250		19-290-3410	WDX138031002250Y	90.20.65-10025OY	VDG20100250		WDX138031002250G
100	315	VDY100315		19-290-3412	WDX138031003315Y		VDG20100315		WDX138031003315G
125	50	VDY125050					VDG20125050		
125	80	VDY125080	W071208	19-290-3503	WDX138031250800Y	90.20.65-12580Y	VDG20125080	W051208	WDX138031250800G
125	100	VDY125100	W071212	19-290-3504	WDX138031251100Y	90.20.65-12510OY	VDG20125100	W051212	WDX138031251100G
125	125	VDY125125	W071212	19-290-3505	WDX138031251125Y	90.20.65-12512SY	VDG20125125	W051212	WDX138031251125G
125	160	VDY125160	W071216	19-290-3506	WDX138031251160Y	90.20.65-12516OY	VDG20125160	W051216	WDX138031251160G
125	200	VDY125200	W071220	19-290-3508	WDX138031252200Y	90.20.65-12520OY	VDG20125200	W051220	WDX138031252200G
125	250	VDY125250		19-290-3510	WDX138031252250Y	90.20.65-12525OY	VDG20125250		WDX138031252250G
125	315	VDY125315		19-290-3512	WDX138031253315Y		VDG20125315		WDX138031253315G
160	50	VDY160050					VDG20160050		
160	80	VDY160080					VDG20160080		
160	100	VDY160100	W071610	19-290-3604	WDX138031601100Y	90.20.65-16010OY	VDG20160100	W051610	WDX138031601100G
160	125	VDY160125	W071612	19-290-3605	WDX138031601125Y	90.20.65-16012SY	VDG20160125	W051612	WDX138031601125G
160	160	VDY160160	W071616	19-290-3606	WDX138031601160Y	90.20.65-16016OY	VDG20160160	W051616	WDX138031601160G
160	200	VDY160200	W071620	19-290-3608	WDX138031602200Y	90.20.65-16020OY	VDG20160200		WDX138031602200G
160	250	VDY160250		19-290-3610	WDX138031602250Y	90.20.65-16025OY	VDG20160250		WDX138031602250G
160	300					90.20.65-16030OY			
160	315	VDY160315		19-290-3612	WDX138031603315Y		VDG20160315		WDX138031603315G
200	200					90.20.65-20030OY			
200	250					90.20.65-20025OY			
200	300					90.20.65-20030OY			

PLEASE NOTE:

- The highlighted items (in grey) have different hole patterns than our existing wear plates. They do not convert to a Lamina part number and must be ordered as highlighted.

Lamina

Automotive Wear Plate Conversions

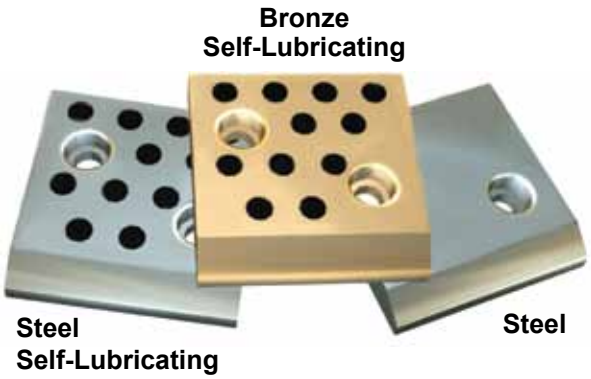
Dimensions		PLAIN STEEL					CAST IRON w/GRAPHITE PLUGS
Width	Length	Anchor Lamina Part Number	NAAMS W06 Series Number	Chrysler Part Number	Ford Part Number Part Number	General Motors Part Number	General Motors Part Number
50	80	VDX050080					
50	100	VDX050100	W060510	19-290-4204	WDX13803050100X	90.20.65-50100X	90.20.65-50100C
50	125	VDX050125	W060512	19-290-4205	WDX13803050125X	90.20.65-50125X	90.20.65-50125C
50	160	VDX050160	W060516	19-290-4206	WDX13803050160X	90.20.65-50160X	90.20.65-50160C
50	200	VDX050200	W060520	19-290-4208	WDX13803050200X	90.20.65-50200X	90.20.65-50200C
80	50	VDX080050					
80	80	VDX080080					
80	100	VDX080100	W060810	19-290-4304	WDX13803080100X	90.20.65-80100X	90.20.65-80100C
80	125	VDX080125	W060812	19-290-4305	WDX13803080125X	90.20.65-80125X	90.20.65-80125C
80	160	VDX080160	W060816	19-290-4306	WDX13803080160X	90.20.65-80160X	90.20.65-80160C
80	200	VDX080200	W060820	19-290-4308	WDX13803080200X	90.20.65-80200X	90.20.65-80200C
80	250	VDX080250		19-290-4310			
80	315	VDX080315		19-290-4312			
100	50	VDX100050		19-290-4402		90.20.65-10050X	90.20.65-10050C
100	80	VDX100080	W061080	19-290-4403	WDX13803100080X	90.20.65-10080X	90.20.65-10080C
100	100	VDX100100	W061010	19-290-4404	WDX13803100100X	90.20.65-100100X	90.20.65-100100C
100	125	VDX100125	W061012	19-290-4405	WDX13803100125X	90.20.65-100125X	90.20.65-100125C
100	160	VDX100160	W061016	19-290-4406	WDX13803100160X	90.20.65-100160X	90.20.65-100160C
100	200	VDX100200	W061020	19-290-4408	WDX13803100200X	90.20.65-100200X	90.20.65-100200C
100	250	VDX100250		19-290-4410	WDX13803100250X	90.20.65-100250X	90.20.65-100250C
100	315	VDX100315		19-290-4412	WDX13803100315X		
125	50	VDX125050					
125	80	VDX125080	W061208	19-290-4503	WDX13803125080X	90.20.65-12580X	90.20.65-12580C
125	100	VDX125100	W061212	19-290-4504	WDX13803125100X	90.20.65-125100X	90.20.65-125100C
125	125	VDX125125	W061212	19-290-4505	WDX13803125125X	90.20.65-125125X	90.20.65-125125C
125	160	VDX125160	W061216	19-290-4506	WDX13803125160X	90.20.65-125160X	90.20.65-125160C
125	200	VDX125200	W061220	19-290-4508	WDX13803125200X	90.20.65-125200X	90.20.65-125200C
125	250	VDX125250		19-290-4510	WDX13803125250X	90.20.65-125250X	90.20.65-125250C
125	315	VDX125315		19-290-4512	WDX13803125315X		
160	50	VDX160050					
160	80	VDX160080					
160	100	VDX160100	W061610	19-290-4604	WDX13803160100X	90.20.65-160100X	90.20.65-160100C
160	125	VDX160125	W061612	19-290-4605	WDX13803160125X	90.20.65-160125X	90.20.65-160125C
160	160	VDX160160	W061616	19-290-4606	WDX13803160160X	90.20.65-160160X	90.20.65-160160C
160	200	VDX160200	W061620	19-290-4608	WDX13803160200X	90.20.65-160200X	90.20.65-160200C
160	250	VDX160250		19-290-4610	WDX13803160250X	90.20.65-160250X	90.20.65-160250C
160	300					90.20.65-160300X	90.20.65-160300C
160	315	VDX160315		19-290-4612	WDX13803160315X		
200	200					90.20.65-200300X	90.20.65-200300C
200	250					90.20.65-200250X	90.20.65-200250C
200	300					90.20.65-200300X	90.20.65-200300C

PLEASE NOTE:

- The highlighted items (in grey) have different hole patterns than our existing wear plates. They do not convert to a Lamina part number and must be ordered as highlighted.

- Cast iron must be ordered using the GM part number.

Wear Plates – NAAMS Series
W01, W02 & W03 – Metric

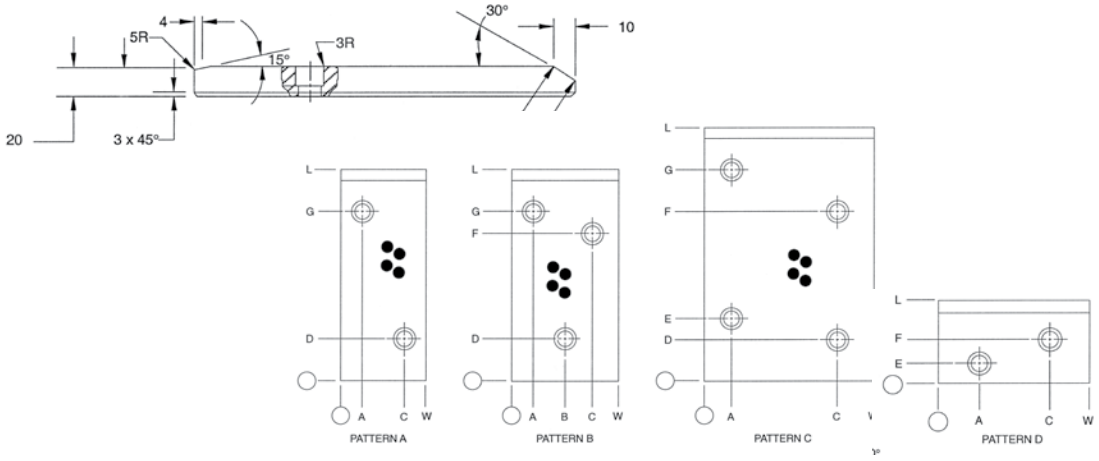


PRODUCT FEATURES

The NAAMS Series of wear plate fully meets the North American Automotive Standard. They are 20mm thick and available in plain steel, steel with graphite plugs and bronze with graphite plugs. The bronze and steel with graphite plugs are self-lubricating products and do not require lubrication during production. All NAAMS wear plates are supplied with mounting holes to accommodate M12 S.H.C.S. For ease of identification, all products are marked with the part number.

Steel Without Graphite Plugs Part #	Steel With Graphite Plugs Part #	Bronze With Graphite Plugs Part #	Width W	Length L
W025010	W015010	W035010	50	100
W025015	W015015	W035015		150
W025020	W015020	W035020		200
W028010	W018010	W038010	80	100
W028015	W018015	W038015		150
W028020	W018020	W038020		200
W028025	W018025	W038025		250
W028031	W018031	W038031		315
W021050	W011050	W031050	100	50
W021080	W011080	W031080		80
W021010	W011010	W031010		100
W021015	W011015	W031015		150
W021020	W011020	W031020		200
W021025	W011025	W031025		250
W021031	W011031	W031031		315
W021280	W011280	W031280	125	80
W021210	W011210	W031210		100
W021215	W011215	W031215		150
W021220	W011220	W031220		200
W021225	W011225	W031225		250
W021231	W011231	W031231		315
W021610	W011610	W031610	160	100
W021615	W011615	W031615		150
W021620	W011620	W031620		200
W021625	W011625	W031625		250
W021631	W011631	W031631		315

Wear Plates – NAAMS Series
W01, W02 & W03 – Metric



MATERIALS:

- Steel wear plates are manufactured from high quality carburized steel. They are precision ground and hardened to Rc 56-60 (1.2 mm to 1.5 mm deep). They are available with or without graphite plugs.
- Bronze wear plates are manufactured from bronze material with oil impregnated graphite plugs.
- When ordered with graphite plugs, the plugs cover 25-30% of the wear surface and are arranged to generate optimal coverage at any engagement direction of the wear plate.

TOLERANCES:

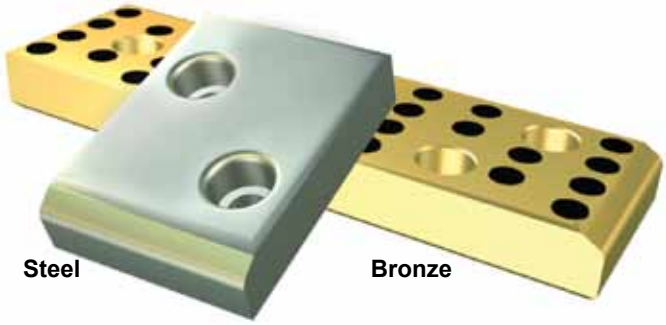
- Thickness ± 0.025 mm
- Mounting hole locations ± 0.25 mm
- Flatness ± 0.012 mm
- Unless otherwise noted ± 0.25 mm

MOUNTING SCREWS:

- M12 S.H.C.S. recommended. Plate has 13.5mm through hole with 20mm counter-bore 13mm deep.

Width W (mm)	Length L (mm)	Mounting Hole Pattern & Location								
		Hole Qty.	Pattern	A (mm)	B (mm)	C (mm)	D (mm)	E (mm)	F (mm)	G (mm)
50	100	2	A	25	–	25	30	–	–	60
	150	2	A		–		30	–	–	110
	200	2	A		–		40	–	–	160
80	100	2	A	20	–	60	30	–	–	60
	150	2	A		–		30	–	–	110
	200	2	A		–		40	–	–	160
	250	2	A		–		40	–	–	210
	315	3	B		40		40	–	250	275
100	50	2	D	22	–	78	–	14	27	–
	80	2	D		–		–	30	50	–
	100	2	A		–		30	–	–	60
	150	2	A		–		30	–	–	110
	200	3	B		50		40	–	135	160
	250	3	B		50		40	–	185	210
	315	3	B		50		40	–	250	275
125	80	2	D	25	–	100	–	30	50	–
	100	2	A		–		30	–	–	60
	150	2	A		–		30	–	–	110
	200	3	B		62		40	–	135	160
	250	3	B		62		40	–	185	210
	315	4	C		–		40	65	250	275
160	100	2	A	30	–	130	30	–	–	60
	150	2	A		–		30	–	–	110
	200	3	B		80		40	–	135	160
	250	4	C		–		40	65	185	210
	315	4	C		–		40	65	250	275

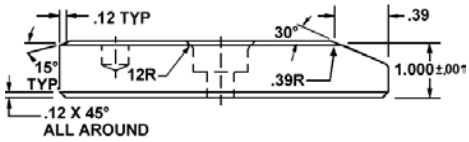
Wear Plates – Automotive – Inch



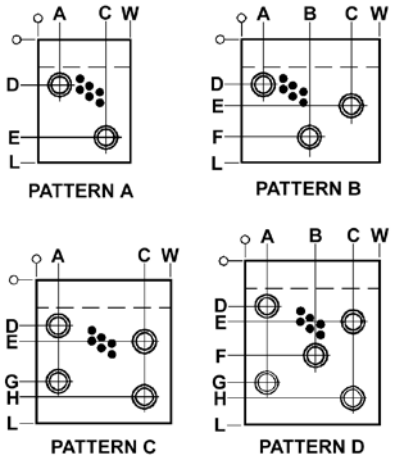
- Steel
- Bronze Self-Lubricating

AVAILABLE FROM STOCK

Lamina offers a range of wear plate styles and sizes for your selection.



WEAR PLATES – INCH													
NSX	NYG	NOSE WIDTH	LENGTH	SCREW HOLES		SCREW HOLE LOCATIONS							
STEEL	BRONZE GRAPHITE	W		NO.	PAT.	A	B	C	D	E	F	G	H
NSX82508	NYG82508	2.50	8.00	2	A	1.25	-	1.25	1.50	6.50	-	-	-
NSX83025	NYG83025	3.00	2.50	2	A	.75	-	2.25	1.25	-	-	-	-
NSX8304	NYG8304	3.00	4.00	2	A	.75	-	2.25	1.25	2.75	-	-	-
NSX8305	NYG8305	3.00	5.00	2	A	.75	-	2.25	1.50	3.75	-	-	-
NSX8306	NYG8306	3.00	6.00	2	A	.75	-	2.25	1.50	4.50	-	-	-
NSX8308	NYG8308	3.00	8.00	2	A	.75	-	2.25	1.50	6.50	-	-	-
NSX8310	NYG8310	3.00	10.00	3	B	.75	1.50	2.25	1.50	2.50	8.50	-	-
NSX8312	NYG8312	3.00	12.00	3	B	.75	1.50	2.25	1.50	2.50	10.50	-	-
NSX84025	NYG84025	4.00	2.50	2	A	1.25	-	2.75	1.25	-	-	-	-
NSX8403	NYG8403	4.00	3.00	2	A	1.00	-	3.00	1.50	2.25	-	-	-
NSX8404	NYG8404	4.00	4.00	2	A	1.00	-	3.00	1.25	2.75	-	-	-
NSX8405	NYG8405	4.00	5.00	2	A	.88	-	3.12	1.50	3.75	-	-	-
NSX8406	NYG8406	4.00	6.00	2	A	.88	-	3.12	1.50	4.50	-	-	-
NSX8408	NYG8408	4.00	8.00	3	B	.88	2.00	3.12	1.50	2.50	6.50	-	-
NSX8410	NYG8410	4.00	10.00	3	B	.88	2.00	3.12	1.50	2.50	8.50	-	-
NSX8412	NYG8412	4.00	12.00	3	B	.88	2.00	3.12	1.50	2.50	10.50	-	-
NSX85025	NYG85025	5.00	2.50	2	A	1.25	-	3.75	1.25	-	-	-	-
NSX8504	NYG8504	5.00	4.00	2	A	1.00	-	4.00	1.50	2.75	-	-	-
NSX8505	NYG8505	5.00	5.00	2	A	1.00	-	4.00	1.50	3.75	-	-	-
NSX8506	NYG8506	5.00	6.00	2	A	1.00	-	4.00	1.50	4.50	-	-	-
NSX8508	NYG8508	5.00	8.00	3	B	1.00	2.50	4.00	1.50	2.50	6.50	-	-
NSX8510	NYG8510	5.00	10.00	3	B	1.00	2.50	4.00	1.50	2.50	8.50	-	-
NSX8512	NYG8512	5.00	12.00	4	C	1.00	-	4.00	1.50	2.50	-	9.50	10.50
NSX8604	NYG8604	6.00	4.00	3	B	1.00	3.00	5.00	1.25	2.25	2.75	-	-
NSX8605	NYG8605	6.00	5.00	2	A	1.12	-	4.88	1.50	3.75	-	-	-
NSX8606	NYG8606	6.00	6.00	4	C	1.00	-	5.00	1.25	2.25	-	3.75	4.75
NSX8608	NYG8608	6.00	8.00	4	C	1.00	-	5.00	1.25	2.25	-	5.75	6.75
NSX8610	NYG8610	6.00	10.00	5	D	1.00	3.00	5.00	1.25	2.25	5.00	7.75	8.75
NSX8612	NYG8612	6.00	12.00	5	D	1.00	3.00	5.00	1.25	2.25	6.00	9.75	10.75
NSX8805	NYG8805	8.00	5.00	2	A	1.25	-	6.75	1.50	3.75	-	-	-
NSX8806	NYG8806	8.00	6.00	4	C	1.00	-	7.00	1.25	2.25	-	3.75	4.75



NOTES & SPECIFICATIONS

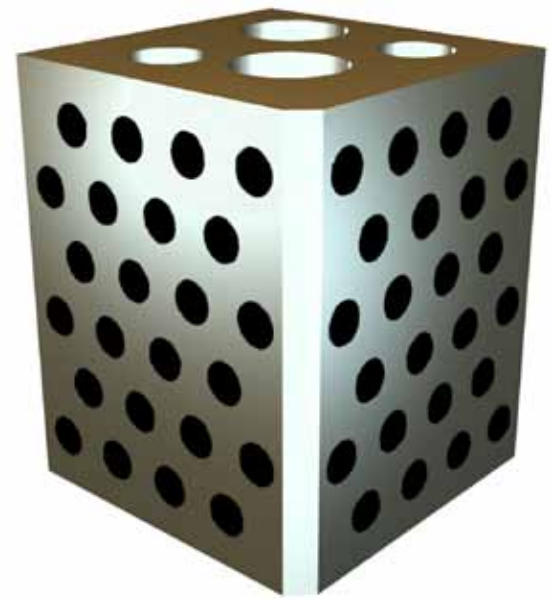
- All dimensions are ±.010 unless noted.
- All screw holes are for 1/2 Soc. Hd. Scr., 9/16 drill through & 13/16 Counterbore x .56 deep.

TOLERANCES:

Thickness	±.001
Hole Location	±.010
Flatness	±.0005

(Do not use grease on graphite-plugged surfaces.)

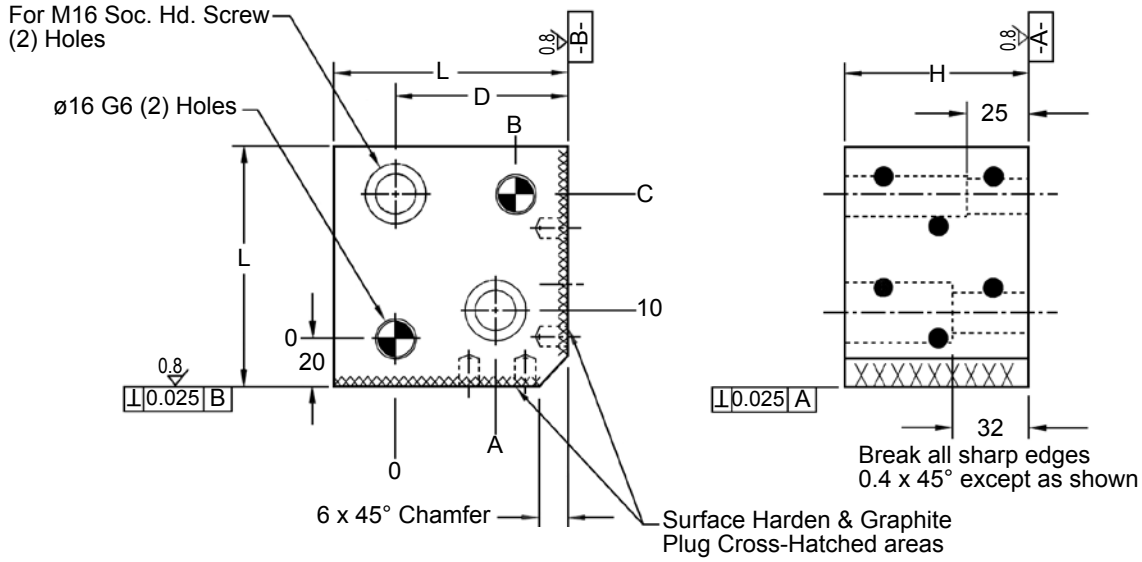
Corner Guide Blocks



- Solid Steel
- Self-Lubricating

AVAILABLE FROM STOCK

AUTOMOTIVE METRIC
NAAMS STANDARD



NOTES & SPECIFICATIONS

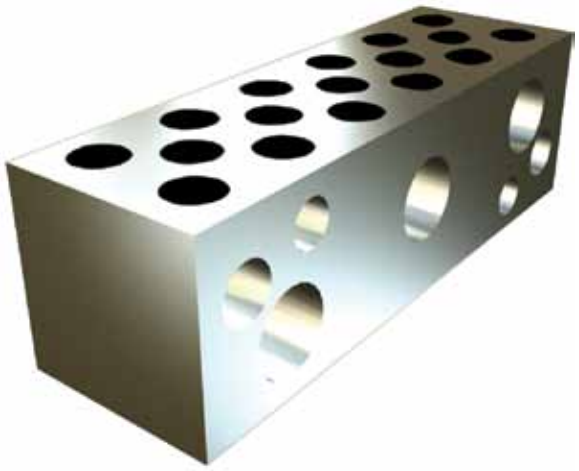
- Screw hole locations ±0.25
- Dowel hole locations ±0.10

Apply a film or mist of light machine oil on new self-lubricating parts before use will enhance the self-lubricating properties.

(Do not use grease on graphite-plugged surfaces.)

CORNER GUIDE BLOCKS – METRIC						
PART NUMBER	H	A	B	C	D	L
		±0.25	±0.013	±0.013	±0.25	
G118007	75	25	35	40	55	80
G118010	100					
G118012	125					
G119507	75	40	50	55	70	95
G119510	100					
G119512	125					
G111207	75	65	75	80	95	120
G111210	100					
G111212	125					

Guide Blocks – Inch



- Solid Steel
- Chromium Steel AISI-4150 or 6150
- Wear area is surface hardened – RC 56-62
- All Blocks are self-lubricating with graphite oil plugs
- All Blocks are precision ground
- All Tolerances not noted are held to ±.010 and are non-cumulative
- Easily altered if necessary
- Lamina logo and part number is etched/stamped on part

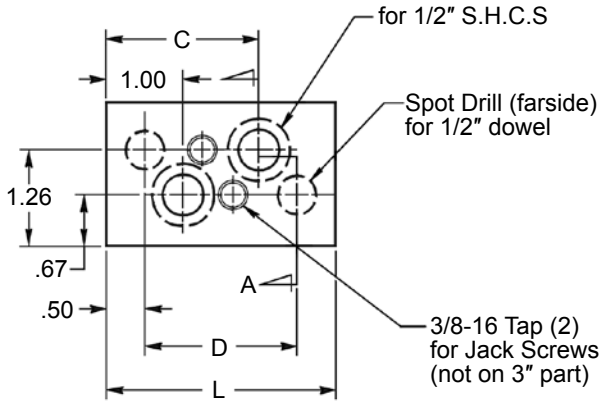
GUIDE BLOCKS – INCH							
PART NUMBER	WIDTH 1.875		A	B	C	D	# MTG. HOLES
	LENGTH	HEIGHT	+.000 –.005				
L14G24G	3.00	1.69	-	-	2.000	2.000	2
L14G32G	4.00		-	-	3.000	3.000	
L14G40G	5.00		-	2.500	4.000	4.000	3
L14G48G	6.00		-	3.000	5.000	5.000	
L14G64G	8.00		2.380	5.620	7.000	7.000	4
L14G80G	10.00		3.500	6.500	9.000	9.000	
L16G24G	3.00	1.94	-	-	2.000	2.000	2
L16G32G	4.00		-	-	3.000	3.000	
L16G40G	5.00		-	2.500	4.000	4.000	3
L16G48G	6.00		-	3.000	5.000	5.000	
L16G64G	8.00		2.380	5.620	7.000	7.000	4
L16G80G	10.00		3.500	6.500	9.000	9.000	
L20G24G	3.00	2.44	-	-	2.000	2.000	2
L20G32G	4.00		-	-	3.000	3.000	
L20G40G	5.00		-	2.500	4.000	4.000	3
L20G48G	6.00		-	3.000	5.000	5.000	
L20G64G	8.00		2.380	5.620	7.000	7.000	4
L20G80G	10.00		3.500	6.500	9.000	9.000	
L24G24G	3.00	2.94	-	-	2.000	2.000	2
L24G32G	4.00		-	-	3.000	3.000	
L24G40G	5.00		-	2.500	4.000	4.000	3
L24G48G	6.00		-	3.000	5.000	5.000	
L24G64G	8.00		2.380	5.620	7.000	7.000	4
L24G80G	10.00		3.500	6.500	9.000	9.000	

Location of mounting holes and dowels are tightly held to provide for interchangeability.

Apply a film or mist of light machine oil on new self-lubricating parts before use will enhance the self-lubricating properties.

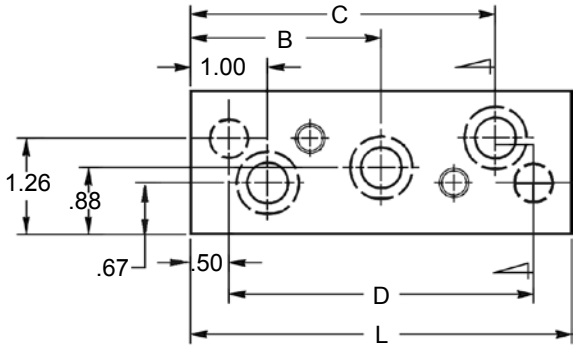
(Do not use grease on graphite-plugged surfaces.)

Guide Blocks – Inch, continued

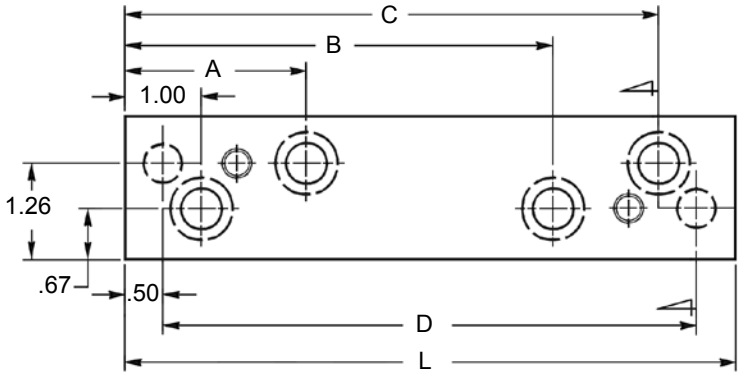


Note: Mounting Hole locations ±.005

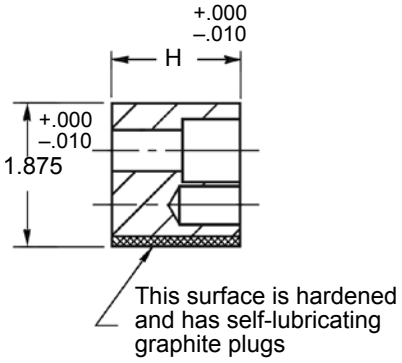
2-HOLE PATTERN



3-HOLE PATTERN



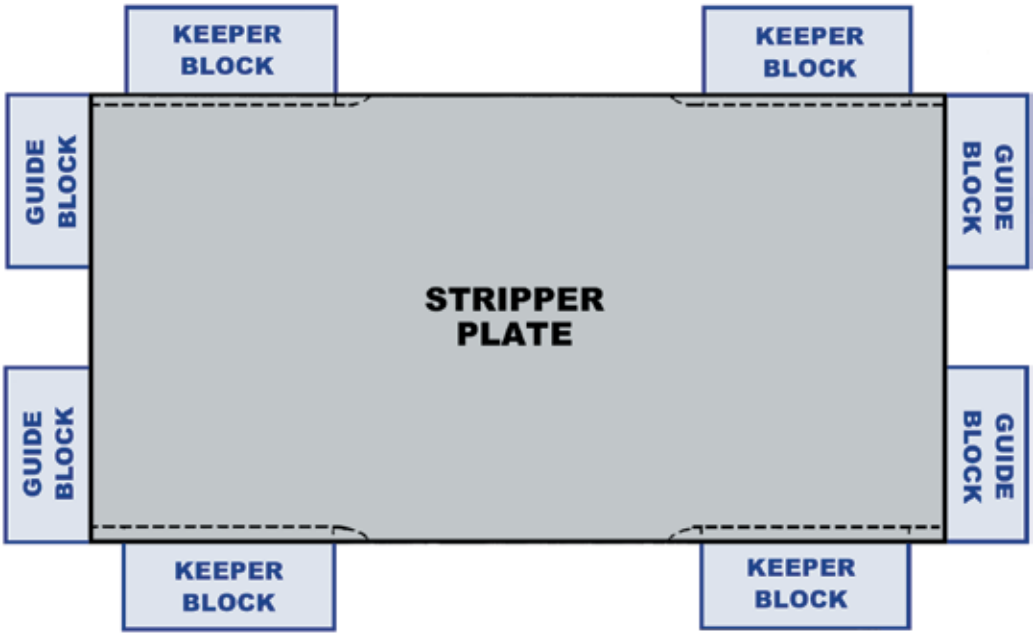
4-HOLE PATTERN



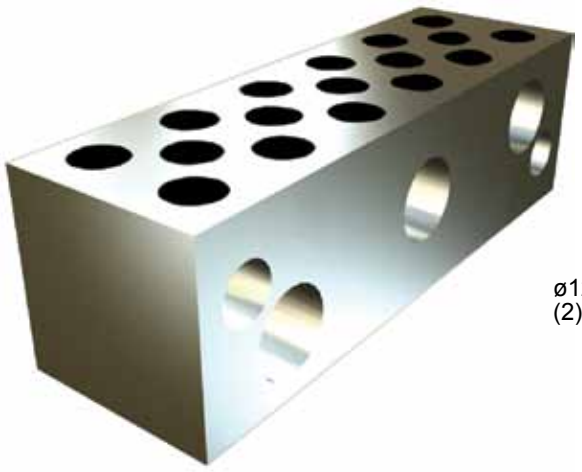
TYPICAL SECTION

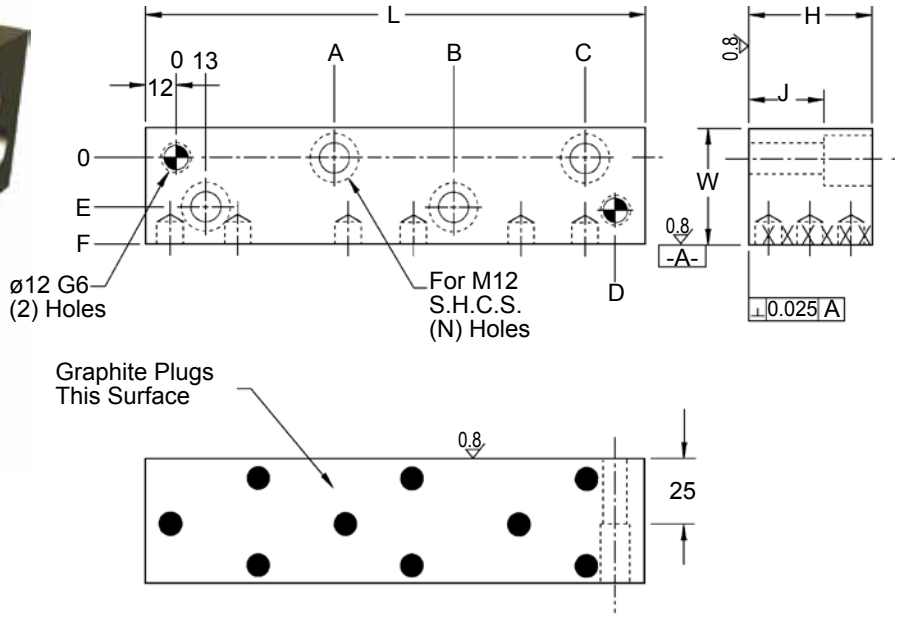
Guide and Keeper Blocks are used together for guidance and alignment.

Keeper Block information starts on page 32.



Guide Blocks – Metric – 45mm Wide





NOTES & SPECIFICATIONS:

- Screw hole locations ±0.25
- Dowel hole locations ±0.10
- Material: Steel
- Cross-hatched areas surface hardened

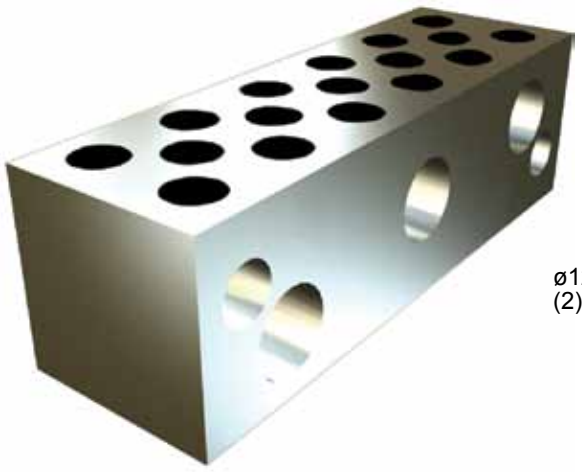
AUTOMOTIVE METRIC NAAMS STANDARD

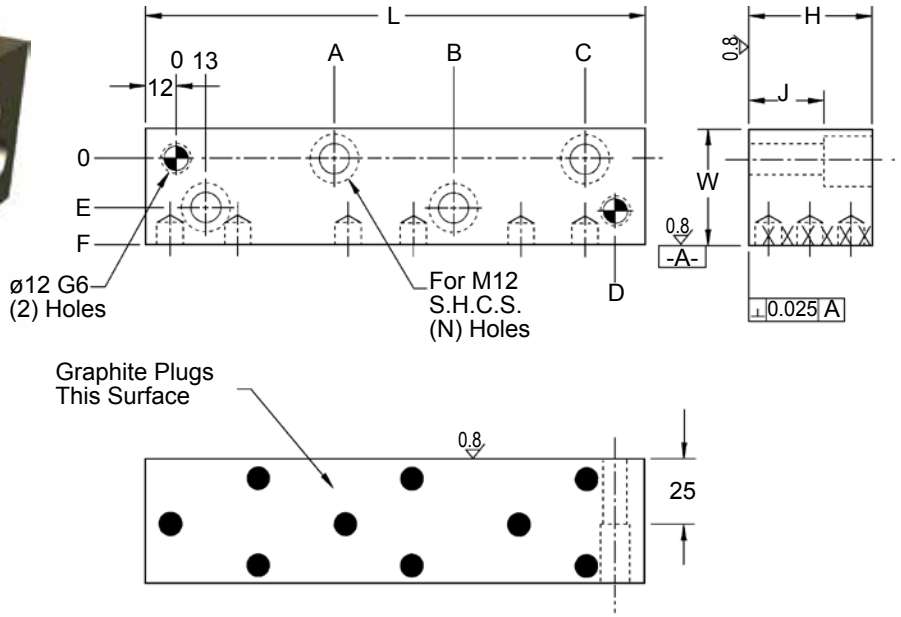
GUIDE BLOCKS – SOLID STEEL – SELF-LUBRICATING – 45mm																							
PART NUMBER	(N) HOLES	L	A	B	C	D	E	F	H	J	W	PART NUMBER	(N) HOLES	L	A	B	C	D	E	F	H	J	W
G014007	2	75	-	-	38	51	18	33	40	15	45	G016007	2	75	-	-	38	51	18	33	60	30	45
G014010	2	100	-	-	63	76						G016010	2	100	-	-	63	76					
G014012	3	125	-	50.5	88	101						G016012	3	125	-	50.5	88	101					
G014015	3	150	-	63.0	113	126						G016015	3	150	-	63.0	113	126					
G014017	3	175	-	75.5	138	151						G016017	3	175	-	75.5	138	151					
G014020	4	200	63	113.0	163	176						G016020	4	200	63	113.0	163	176					
G014022	4	225	73	128.0	188	201						G016022	4	225	73	128.0	188	201					
G014025	4	250	83	153.0	213	226						G016025	4	250	83	153.0	213	226					
G014507	2	75	-	-	38	51	18	33	45	25	45	G017507	2	75	-	-	38	51	18	33	75	30	45
G014510	2	100	-	-	63	76						G017510	2	100	-	-	63	76					
G014512	3	125	-	50.5	88	101						G017512	3	125	-	50.5	88	101					
G014515	3	150	-	63.0	113	126						G017515	3	150	-	63.0	113	126					
G014517	3	175	-	75.5	138	151						G017517	3	175	-	75.5	138	151					
G014520	4	200	63	113.0	163	176						G017520	4	200	63	113.0	163	176					
G014522	4	225	73	128.0	188	201						G017522	4	225	73	128.0	188	201					
G014525	4	250	83	153.0	213	226						G017525	4	250	83	153.0	213	226					
G015007	2	75	-	-	38	51	18	33	50	30	45												
G015010	2	100	-	-	63	76																	
G015012	3	125	-	50.5	88	101																	
G015015	3	150	-	63.0	113	126																	
G015017	3	175	-	75.5	138	151																	
G015020	4	200	63	113.0	163	176																	
G015022	4	225	73	128.0	188	201																	
G015025	4	250	83	153.0	213	226																	

Applying a film or mist of light machine oil on new self-lubricating parts before use will enhance the self-lubricating properties.

(Do not use grease on graphite-plugged surfaces.)

Guide Blocks – Metric – 60mm Wide





NOTES & SPECIFICATIONS:

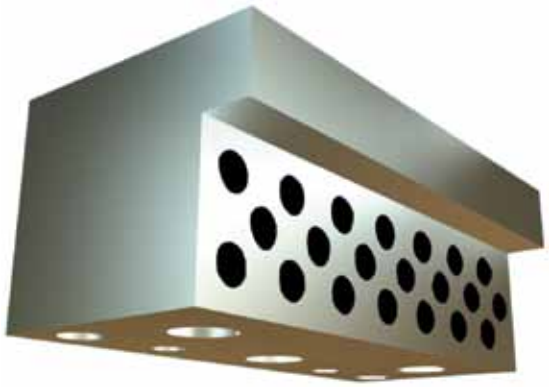
- Screw hole locations ±0.25
- Dowel hole locations ±0.10
- Material: Steel
- Cross-hatched areas surface hardened

AUTOMOTIVE METRIC NAAMS STANDARD

GUIDE BLOCKS – SOLID STEEL – SELF-LUBRICATING – 60mm																						
PART NUMBER	(N) HOLES	L	A	B	C	D	E	F	H	J	W	PART NUMBER	(N) HOLES	L	A	B	C	D	E	F	H	J	W
G025007	2	75	-	-	38	51	33	48	50	30	60	G028007	2	75	-	-	38	51	33	48	80	30	60
G025010	2	100	-	-	63	76						G028010	2	100	-	-	63	76					
G025012	3	125	-	50.5	88	101						G028012	3	125	-	50.5	88	101					
G025015	3	150	-	63.0	113	126						G028015	3	150	-	63.0	113	126					
G025017	3	175	-	75.5	138	151						G028017	3	175	-	75.5	138	151					
G025020	4	200	63	113.0	163	176						G028020	4	200	200	113.0	163	176					
G025022	4	225	73	128.0	188	201						G028022	4	225	225	128.0	188	201					
G025025	4	250	83	153.0	213	226						G025025	4	250	250	153.0	213	226					
G026007	2	75	-	-	38	51	33	48	60	30	60	G020007	2	75	-	-	38	51	33	48	100	30	60
G026010	2	100	-	-	63	76						G020010	2	100	-	-	63	76					
G026012	3	125	-	50.5	88	101						G020012	3	125	-	50.5	88	101					
G026015	3	150	-	63.0	113	126						G020015	3	150	-	63.0	113	126					
G026017	3	175	-	75.5	138	151						G020017	3	175	-	75.5	138	151					
G026020	4	200	63	113.0	163	176						G020020	4	200	200	113.0	163	176					
G026022	4	225	73	128.0	188	201						G020022	4	225	225	128.0	188	201					
G026025	4	250	83	153.0	213	226						G020025	4	250	250	153.0	213	226					

Please check our website – www.anchorlamina.com – for the latest technical information.

Keeper Blocks – Light-Duty



- **Solid Steel**
- **Chromium Steel AISI-4150 or 6150**
- **Wear area is surface hardened – RC 56-62**
- **All Blocks are self-lubricating with graphite oil plugs**
- **All Blocks are precision ground**
- **All Tolerances not noted are held to ±.010 and are non-cumulative**
- **Easily altered if necessary**
- **Lamina logo and part number is etched/stamped on part**

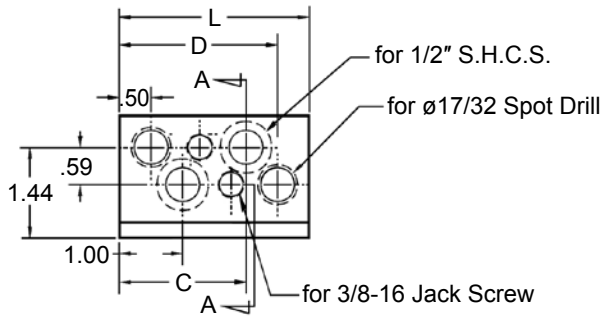
Apply a film or mist of light machine oil on new self-lubricating parts before use will enhance the self-lubricating properties.

(Do not use grease on graphite-plugged surfaces.)

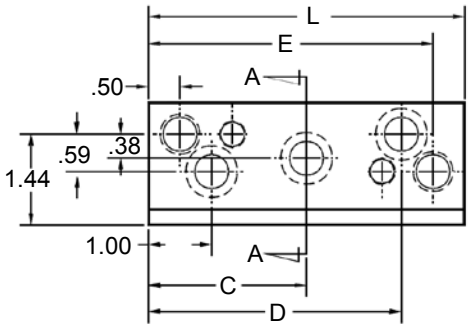
GENERAL DIMENSIONS				
L	C	D	E	F
	±.005			
3.00	2.000	2.500	-	-
4.00	3.000	3.500	-	-
5.00	2.500	4.000	4.500	-
6.00	3.000	5.000	5.500	-
8.00	2.380	5.620	7.000	7.500
10.00	3.500	6.500	9.000	9.500

KEEPER BLOCKS – LIGHT DUTY – SOLID STEEL – SELF-LUBRICATING											
PART NUMBER	L	H	J	K	# MTG. HOLES	PART NUMBER	L	H	J	K	# MTG. HOLES
		±.005	±.001					±.005	±.001		
L8K24G	3.00	1.44	1.000	0.75	2	L12K24G	3.00	1.94	1.500	1.00	2
L8K32G	4.00				2	L12K32G	4.00				2
L8K40G	5.00				3	L12K40G	5.00				3
L8K48G	6.00				3	L12K48G	6.00				3
L8K64G	8.00				4	L12K64G	8.00				4
L8K80G	10.00				4	L12K80G	10.00				4
L10K24G	3.00	1.69	1.250	1.00	2	L14K24G	3.00	2.44	1.750	1.25	2
L10K32G	4.00				2	L14K32G	4.00				2
L10K40G	5.00				3	L14K40G	5.00				3
L10K48G	6.00				3	L14K48G	6.00				3
L10K64G	8.00				4	L14K64G	8.00				4
L10K80G	10.00				4	L14K80G	10.00				4

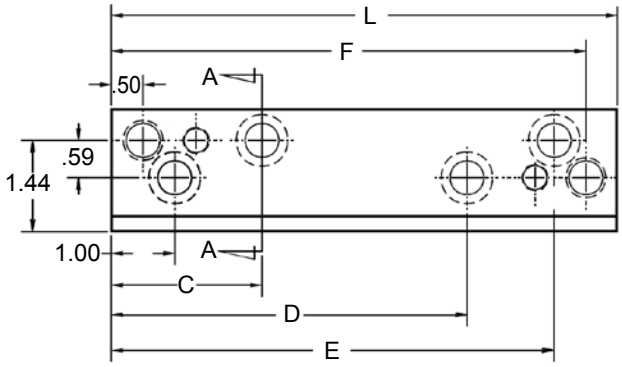
Keeper Blocks – Light-Duty, continued



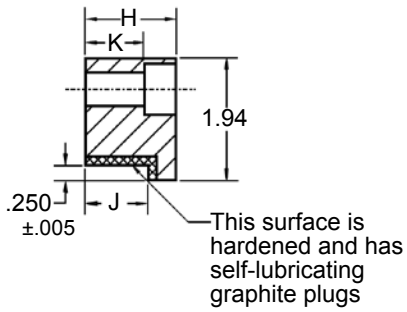
FOR 3 AND 4 INCH LENGTHS



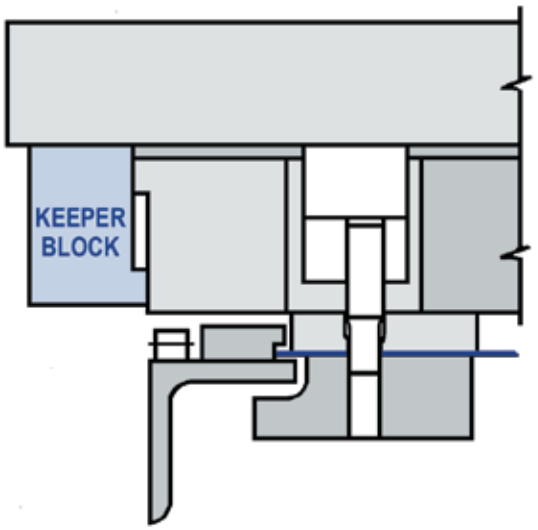
FOR 5 AND 6 INCH LENGTHS



FOR 8 TO 10 INCH LENGTHS

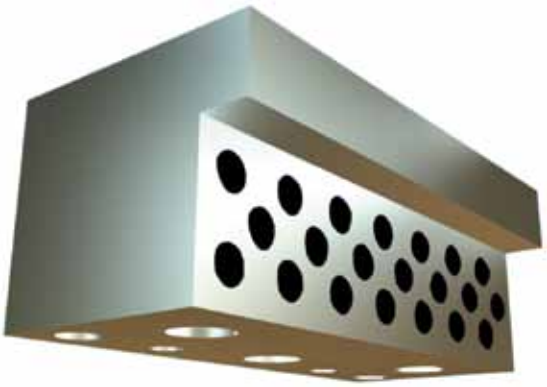


TYPICAL SECTION A–A



Keeper Block as Side Guide

Keeper Blocks – Medium-Duty



- Solid Steel
- Chromium Steel AISI-4150 or 6150
- Wear area is surface hardened – RC 56-62
- All Blocks are self-lubricating with graphite oil plugs
- All Blocks are precision ground
- All Tolerances not noted are held to ±.010 and are non-cumulative
- Easily altered if necessary
- Lamina logo and part number is etched/stamped on part

Apply a film or mist of light machine oil on new self-lubricating parts before use will enhance the self-lubricating properties.

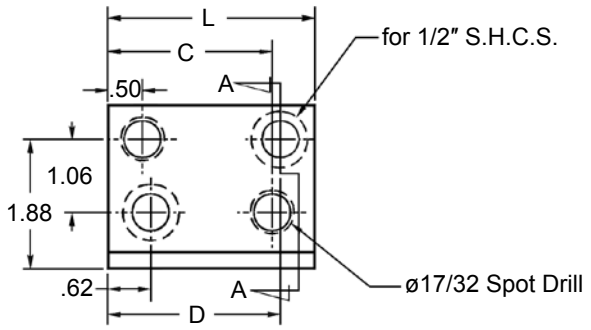
(Do not use grease on graphite-plugged surfaces.)

GENERAL DIMENSIONS					
L	C	D	E	F	G
	±.005				
3.00	2.380	2.500	-	-	-
4.00	2.120	3.380	3.500	-	-
5.00	2.120	4.380	4.500	-	-
6.00	2.120	5.380	5.500	-	-
8.00	2.120	5.880	7.380	7.500	-
10.00	2.120	5.380	7.880	9.380	9.500

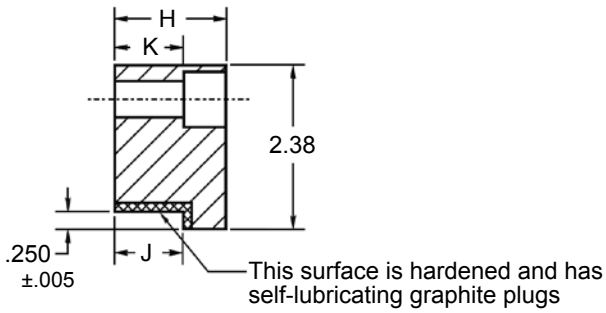
KEEPER BLOCKS – MEDIUM DUTY – SOLID STEEL – SELF-LUBRICATING – INCH

PART NUMBER	L	H ±.005	J ±.001	K	# MTG. HOLES	PART NUMBER	L	H ±.005	J ±.001	K	# MTG. HOLES
L8KM24G	3.00	1.62	1.000	1.00	2	L16KM24G	3.00	2.69	2.000	1.75	2
L8KM32G	4.00				3	L16KM32G	4.00				3
L8KM40G	5.00				3	L16KM40G	5.00				3
L8KM48G	6.00				3	L16KM48G	6.00				3
L8KM64G	8.00				4	L16KM64G	8.00				4
L8KM80G	10.00				5	L16KM80G	10.00				5
L10KM24G	3.00	1.88	1.250	1.25	2	L18KM24G	3.00	2.88	2.250	1.75	2
L10KM32G	4.00				3	L18KM32G	4.00				3
L10KM40G	5.00				3	L18KM40G	5.00				3
L10KM48G	6.00				3	L18KM48G	6.00				3
L10KM64G	8.00				4	L18KM64G	8.00				4
L10KM80G	10.00				5	L18KM80G	10.00				5
L12KM24G	3.00	2.12	1.500	1.50	2	L20KM24G	3.00	3.25	2.500	1.75	2
L12KM32G	4.00				3	L20KM32G	4.00				3
L12KM40G	5.00				3	L20KM40G	5.00				3
L12KM48G	6.00				3	L20KM48G	6.00				3
L12KM64G	8.00				4	L20KM64G	8.00				4
L12KM80G	10.00				5	L20KM80G	10.00				5
L14KM24G	3.00	2.44	1.750	1.75	2	L22KM24G	3.00	3.38	2.750	1.63	2
L14KM32G	4.00				3	L22KM32G	4.00				3
L14KM40G	5.00				3	L22KM40G	5.00				3
L14KM48G	6.00				3	L22KM48G	6.00				3
L14KM64G	8.00				4	L22KM64G	8.00				4
L14KM80G	10.00				5	L22KM80G	10.00				5

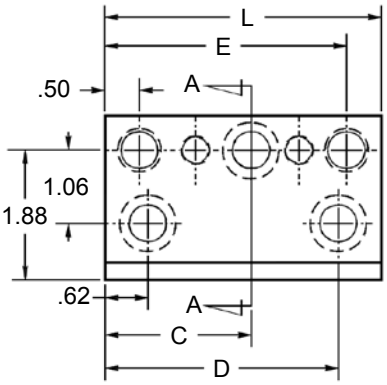
Keeper Blocks – Medium-Duty, continued



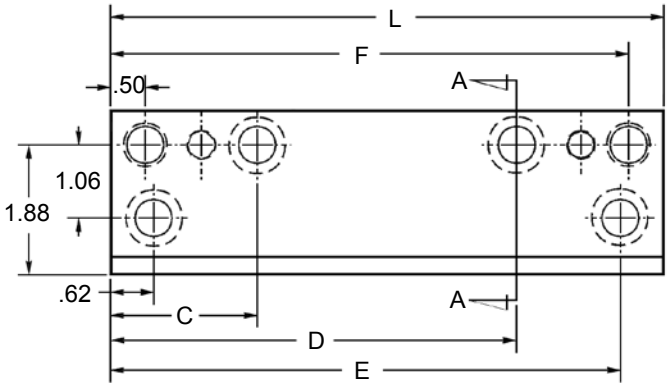
For 3 Inch Length



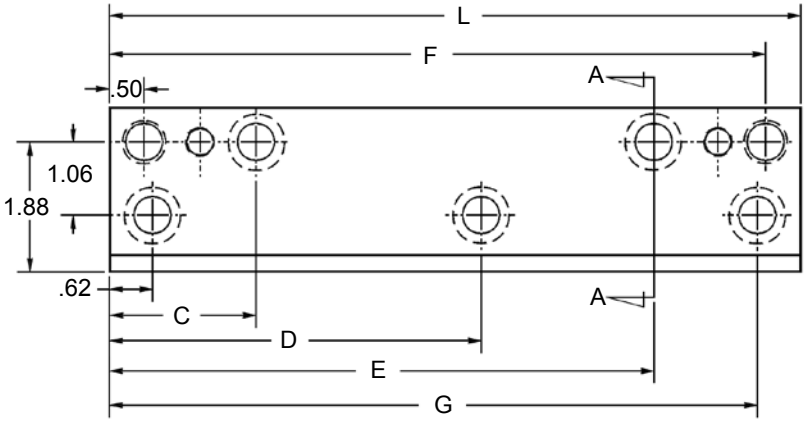
Typical Section A–A



For 4 to 6 Inch Length

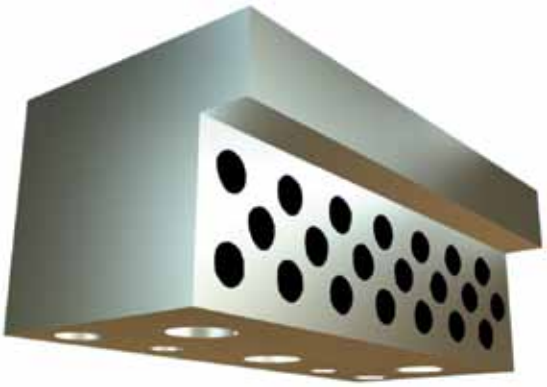


For 8 Inch Length



For 10 Inch Length

Keeper Blocks – Heavy-Duty



- Solid Steel
- Chromium Steel AISI-4150 or 6150
- Wear area is surface hardened – RC 56-62
- All Blocks are self-lubricating with graphite oil plugs
- All Blocks are precision ground
- All Tolerances not noted are held to ±.010 and are non-cumulative
- Easily altered if necessary
- Lamina logo and part number is etched/stamped on part

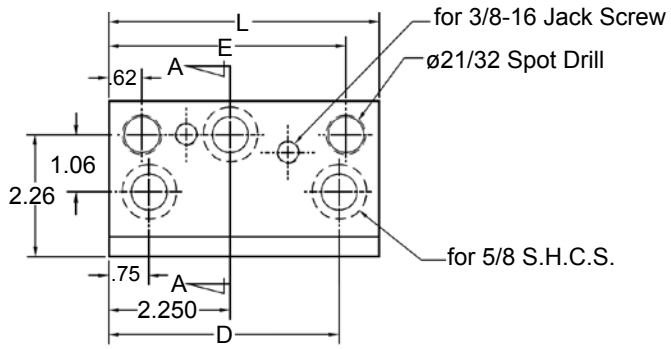
Apply a film or mist of light machine oil on new self-lubricating parts before use will enhance the self-lubricating properties.

(Do not use grease on graphite-plugged surfaces.)

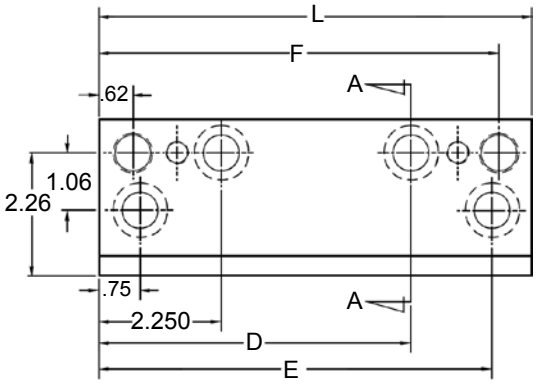
GENERAL DIMENSIONS				
L	D	E	F	G
	±.005			
4.00	3.250	3.380	-	-
5.00	4.250	4.380	-	-
6.00	5.250	5.380	-	-
8.00	5.750	7.250	7.380	-
10.00	5.250	7.750	9.250	9.380
12.00	5.750	9.750	11.250	11.380

KEEPER BLOCKS – HEAVY DUTY – SOLID STEEL – SELF-LUBRICATING – INCH											
PART NUMBER	L	H ±.005	J ±.001	K	# MTG. HOLES	PART NUMBER	L	H ±.005	J ±.001	K	# MTG. HOLES
L14KH32G	4.00	2.62	1.750	1.50	3	L20KH32G	4.00	3.38	2.500	1.75	3
L14KH40G	5.00				3	L20KH40G	5.00				3
L14KH48G	6.00				3	L20KH48G	6.00				3
L14KH64G	8.00				4	L20KH64G	8.00				4
L14KH80G	10.00				5	L20KH80G	10.00				5
L14KH96G	12.00				5	L20KH96G	12.00				5
L16KH32G	4.00	2.88	2.000	1.50	3	L22KH32G	4.00	3.62	2.750	2.25	3
L16KH40G	5.00				3	L22KH40G	5.00				3
L16KH48G	6.00				3	L22KH48G	6.00				3
L16KH64G	8.00				4	L22KH64G	8.00				4
L16KH80G	10.00				5	L22KH80G	10.00				5
L16KH96G	12.00				5	L22KH96G	12.00				5
L18KH32G	4.00	3.12	2.250	1.50	3	L24KH32G	4.00	3.88	3.000	2.25	3
L18KH40G	5.00				3	L24KH40G	5.00				3
L18KH48G	6.00				3	L24KH48G	6.00				3
L18KH64G	8.00				4	L24KH64G	8.00				4
L18KH80G	10.00				5	L24KH80G	10.00				5
L18KH96G	12.00				5	L24KH96G	12.00				5

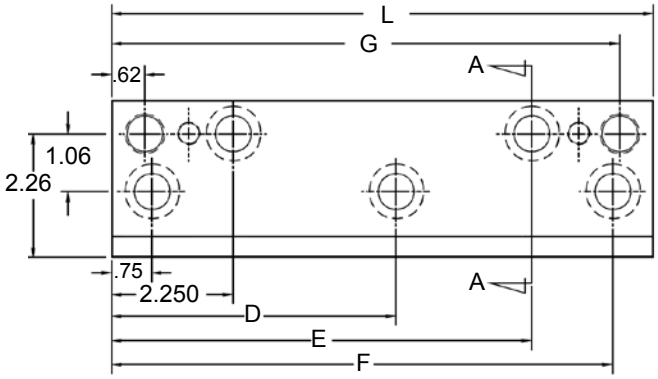
Keeper Blocks – Heavy-Duty, continued



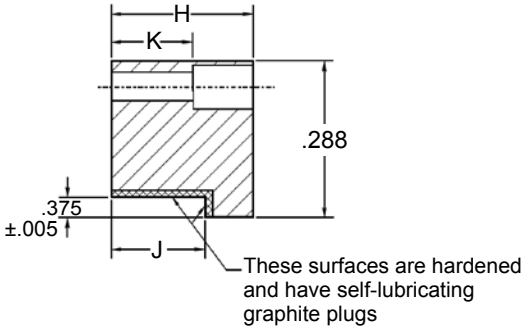
For 4 to 6 Inch Length



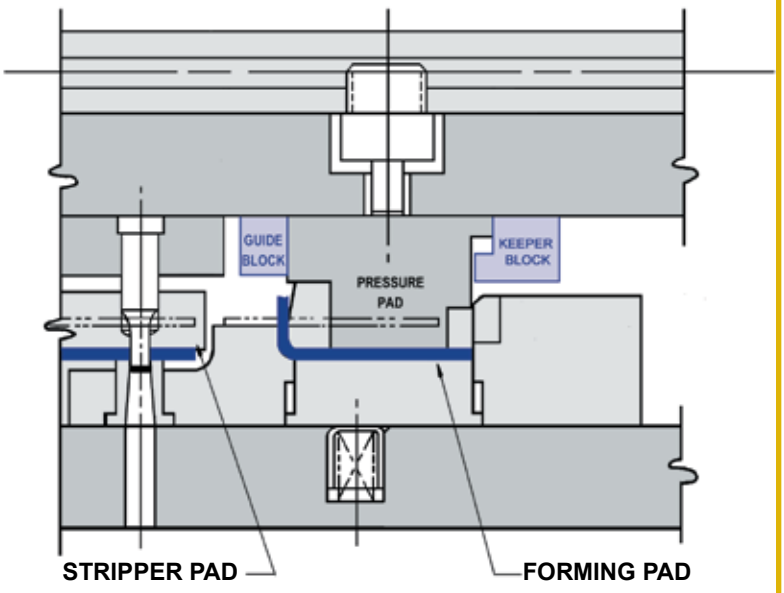
For 8 Inch Length



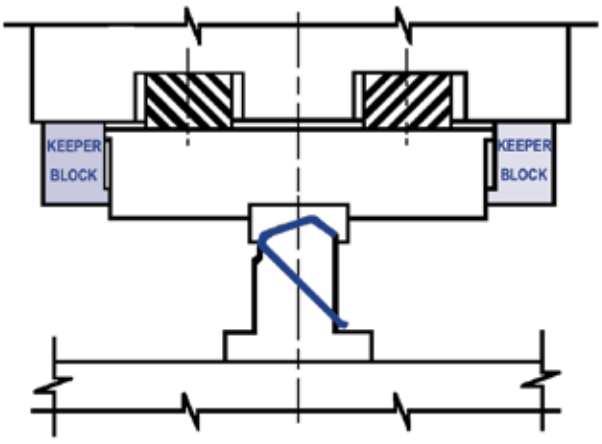
For 10 & 12 Inch Length



Typical Section A-A

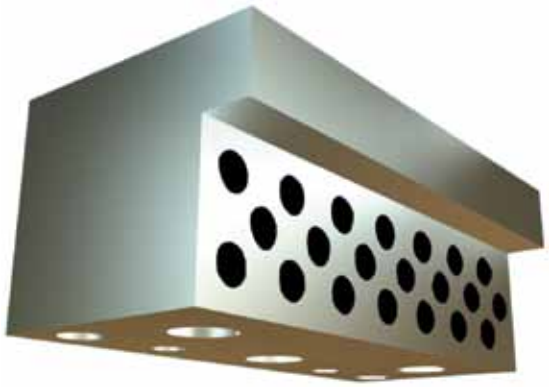


Keeper Blocks help keep the load aligned for a smooth running die operation.

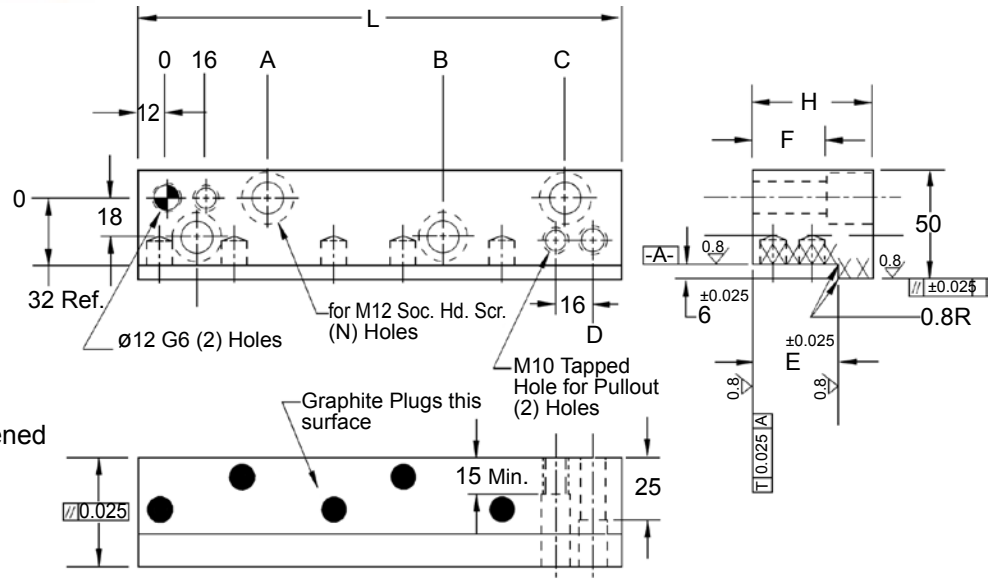


Keeper Blocks – 6mm

AUTOMOTIVE METRIC NAAMS STANDARD



- **Solid Steel**
- **Chromium Steel AISI-4150 or 6150**
- **Wear area is surface hardened – RC 56-62**
- **All Blocks are self-lubricating with graphite oil plugs**
- **All Blocks are precision ground**
- **All Tolerances not noted are held to ±.010 and are non-cumulative**
- **Easily altered if necessary**
- **Lamina logo and part number is etched/stamped on part**



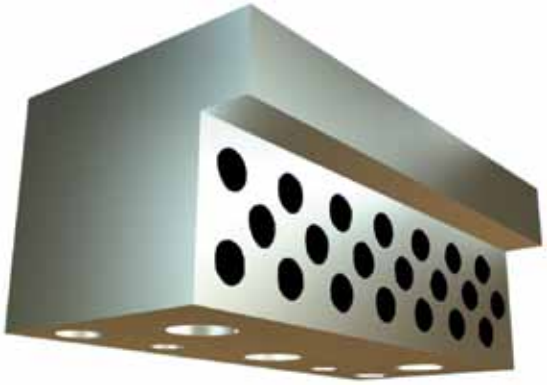
NOTES & SPECIFICATIONS:

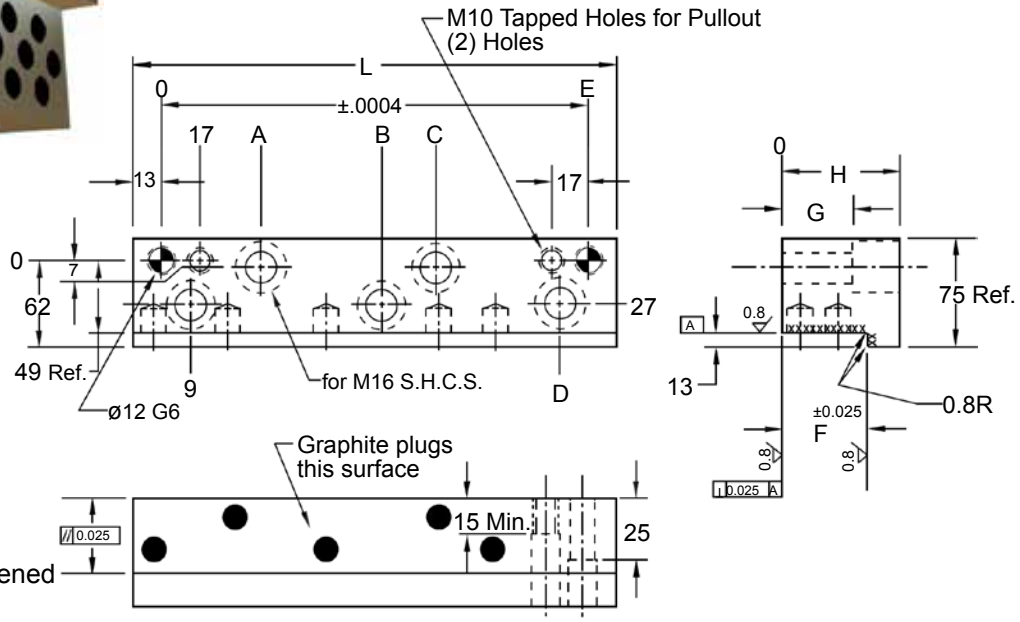
- Screw hole locations ±0.25
- Dowel hole locations ±0.10
- Material: Steel
- Cross-hatched areas surface hardened

KEEPER BLOCKS – SOLID-STEEL – SELF-LUBRICATING – 6mm									
PART NUMBER	(N) HOLES	L	A	B	C	D	E	F	H
			±0.25				±0.025		
R012507	2	75	-	-	38	51	25	15	37
R012510	2	100	-	-	63	76			
R012515	3	150	-	63	113	126			
R012520	4	200	63	113	163	176			
R013207	2	75	-	-	38	51	32	15	45
R013210	2	100	-	-	63	76			
R013215	3	150	-	63	113	126			
R013220	4	200	63	113	163	176			
R013707	2	75	-	-	38	51	37	30	50
R013710	2	100	-	-	63	76			
R013715	3	150	-	63	113	126			
R013720	4	200	63	113	163	176			
R014810	2	100	-	-	63	76	48	30	60
R014815	3	100	-	63	113	126			
R014820	4	150	63	113	163	176			
R014825	4	200	63	113	213	226			

Keeper Blocks – 13mm

AUTOMOTIVE METRIC
NAAMS STANDARD



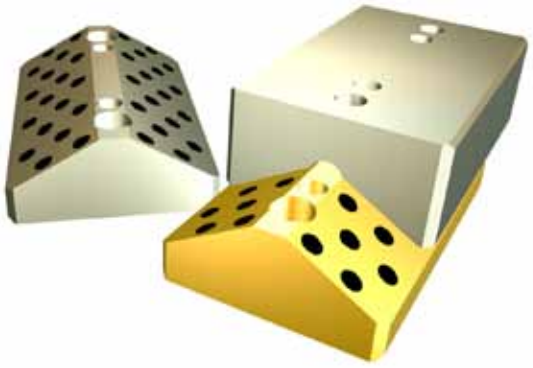


NOTES & SPECIFICATIONS:

- Screw hole locations ±0.25
- Dowel hole locations ±0.10
- Material: Steel
- Cross-hatched areas surface hardened

KEEPER BLOCKS – SOLID-STEEL – SELF-LUBRICATING – 13mm																					
PART NUMBER	(N) HOLES	L	A	B	C	D	E	F	G	H	PART NUMBER	(N) HOLES	L	A	B	C	D	E	F	G	H
			±0.25				±0.025							±0.25				±0.025			
R023510	2	100	-	-	-	65	74				R025510	2	100	-	-	-	65	74			
R023512	3	125	49.5	-	-	90	99				R025512	3	125	49.5	-	-	90	99			
R023515	3	150	62.0	-	-	115	124				R025515	3	150	62.0	-	-	115	124			
R023520	4	200	62.0	-	112	165	174	35	25	50	R025520	4	200	62.0	-	112	165	174	55	35	80
R023522	5	225	62.0	99.5	137	190	199				R025522	5	225	62.0	99.5	137	190	199			
R023525	5	250	62.0	112.0	162	215	224				R025525	5	250	62.0	112.0	162	215	224			
R023530	5	300	62.0	137.0	212	265	274				R025530	5	300	62.0	137.0	212	265	274			
R024010	2	100	-	-	-	65	74				R026510	2	100	-	-	-	65	74			
R024012	3	125	49.5	-	-	90	99				R026512	3	125	49.5	-	-	90	99			
R024015	3	150	62.0	-	-	115	124				R026515	3	150	62.0	-	-	115	124			
R024020	4	200	62.0	-	112	165	174	40	25	60	R026520	4	200	62.0	-	112	165	174	65	45	90
R024022	5	225	62.0	99.5	137	190	199				R026522	5	225	62.0	99.5	137	190	199			
R024025	5	250	62.0	112.0	162	215	224				R026525	5	250	62.0	112.0	162	215	224			
R024030	5	300	62.0	137.0	212	265	274				R026530	5	300	62.0	137.0	212	265	274			
R025010	2	100	-	-	-	65	74				R027510	2	100	-	-	-	65	74			
R025012	3	125	49.5	-	-	90	99				R027512	3	125	49.5	-	-	90	99			
R025015	3	150	62.0	-	-	115	124				R027515	3	150	62.0	-	-	115	124			
R025020	4	200	62.0	-	112	165	174	50	30	70	R027520	4	200	62.0	-	112	165	174	75	45	100
R025022	5	225	62.0	99.5	137	190	199				R027522	5	225	62.0	99.5	137	190	199			
R025025	5	250	62.0	112.0	162	215	224				R027525	5	250	62.0	112.0	162	215	224			
R025030	5	300	62.0	137.0	212	265	274				R027530	5	300	62.0	137.0	212	265	274			

NAAMS U & V Blocks – Metric



PRODUCT FEATURES

These 65, 75, 125 and 175mm U and V Blocks are manufactured to metric dimensions and fully meet the NAAMS standard. The U Block is manufactured from hardened steel. The V block is self-lubricating in design and is available in either bronze with oil-impregnated graphite plugs or steel with oil-impregnated graphite plugs.

FIGURE A - Steel U Block

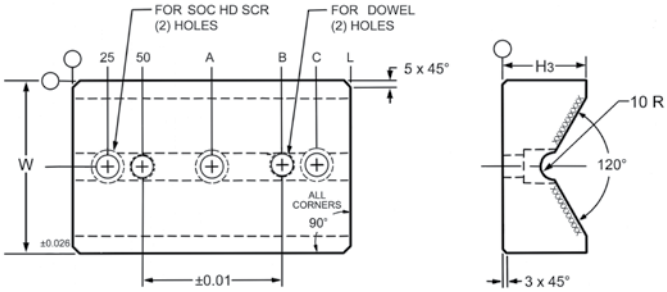


FIGURE B - Self-lubricating V Block

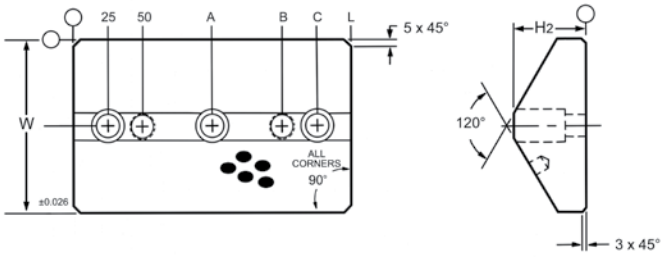
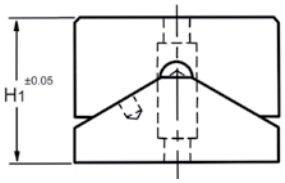


FIGURE C - Assembly



MATERIALS:

- U Blocks are manufactured from high quality carburized steel. They are precision ground and hardened to Rc56-62.
- V Blocks are manufactured from bronze material with oil impregnated graphite plugs or steel with oil-impregnated graphite plugs.

TOLERANCES:

- Between screw holes ± 0.12mm
- Between dowel holes ± 0.10mm
- Unless otherwise noted ± 0.25mm

RECOMMENDED MOUNTING SCREWS & DOWELS:

- **65mm blocks:** M12 S.H.C.S./M12 DOWELS
- **75mm blocks:** M12 S.H.C.S. /M12 DOWELS
- **125mm blocks:** M16 S.H.C.S. M16 DOWELS
- **175mm blocks:** M16 S.H.C.S. M16 DOWELS

NAAMS U & V Blocks – Metric

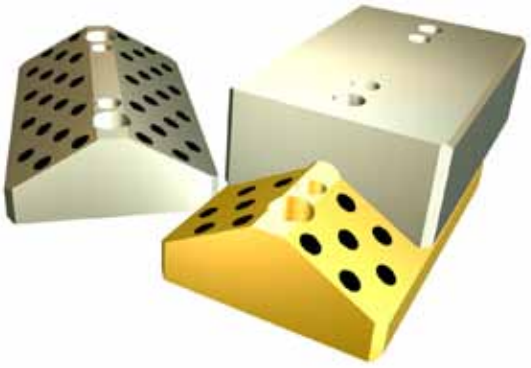
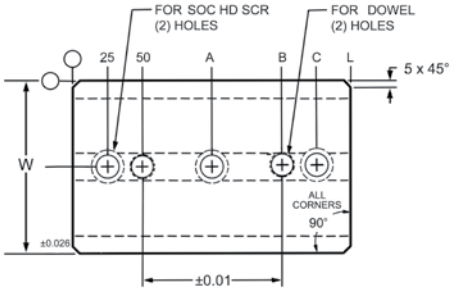


FIGURE A - Steel U Block



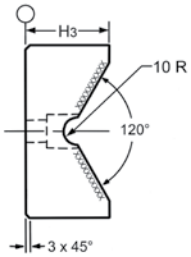
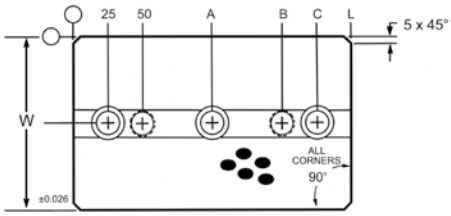


FIGURE B - Self-lubricating V Block



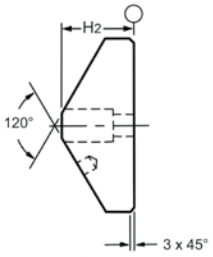
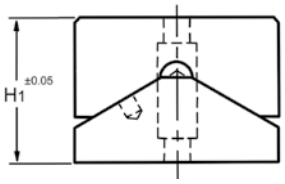


FIGURE C - Assembly



U Block	Steel V Block	Bronze V Block	Width W	Length L	Hole Qty.	A	B	C	H1	H2	H3
Part Number	Part Number	Part Number	(mm)	(mm)		(mm)	(mm)	(mm)	(mm)	(mm)	(mm)
G157515	G167515	G177515	65	150	2	–	100	125	65	39	40
G157520	G167520	G177520	65	200	2	–	150	175	65	39	40
G157525	G167525	G177525	65	250	3	125	200	225	65	39	40
G157530	G167530	G177530	65	300	3	150	250	275	65	39	40
G217515	G227515	G237515	75	150	2	–	100	125	65	39	40
G217520	G227520	G237520	75	200	2	–	150	175	65	39	40
G217525	G227525	G237525	75	250	3	125	200	225	65	39	40
G217530	G227530	G237530	75	300	3	150	250	275	65	39	40
G311215	G321215	G331215	125	150	2	–	100	125	85	52	60
G311220	G321220	G331220	125	200	2	–	150	175	85	52	60
G311225	G321225	G331225	125	250	3	125	200	225	85	52	60
G311230	G321230	G331230	125	300	3	150	250	275	85	52	60
G341715	G351715	G361715	175	150	2	–	100	125	110	67	80
G341720	G351720	G361720	175	200	2	–	150	175	110	67	80

Center Keys & Guides

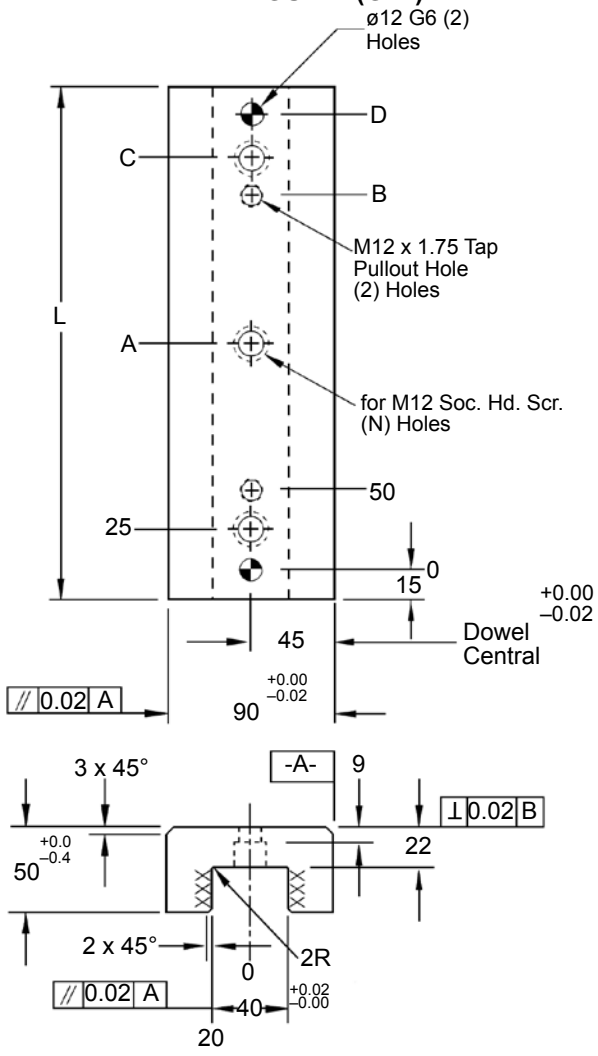
AUTOMOTIVE METRIC
NAAMS STANDARD



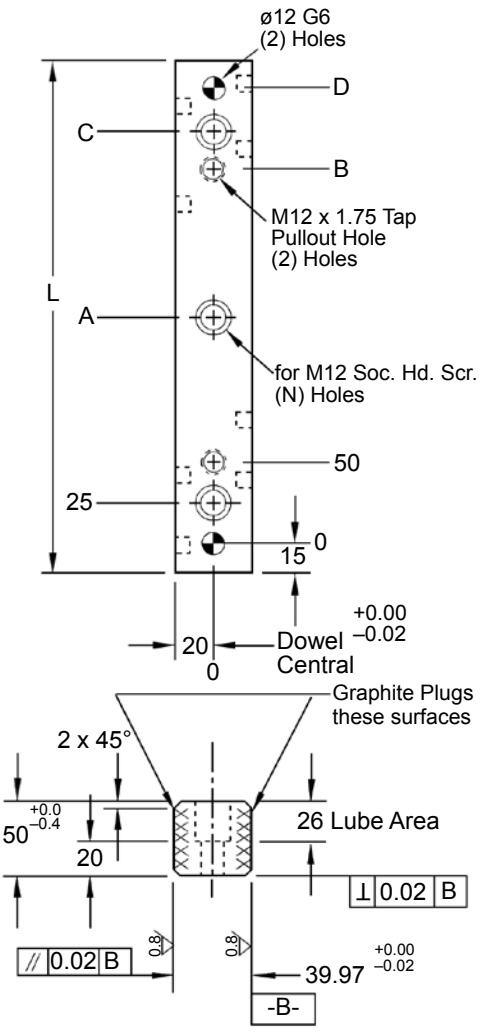
X Steel

Y Bronze Self-Lubricating

CENTER KEY GUIDE (G41)



CENTER KEY (G42 & 43)



CENTER KEYS & GUIDES – METRIC							
NAAMS CODE (X)	NAAMS CODE (Y)	A	B	C	D	L	"N"
G419020	–	-	120	145	170	200	2
G419030	–	135	220	245	270	300	3
G424020	G434020	-	120	145	170	200	2
G424030	G434030	135	220	245	270	300	3

NOTES & SPECIFICATIONS:

- Screw hole locations ±0.25
- Dowel hole locations ±0.10
- Material: Steel
- Cross-hatched areas surface hardened

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